

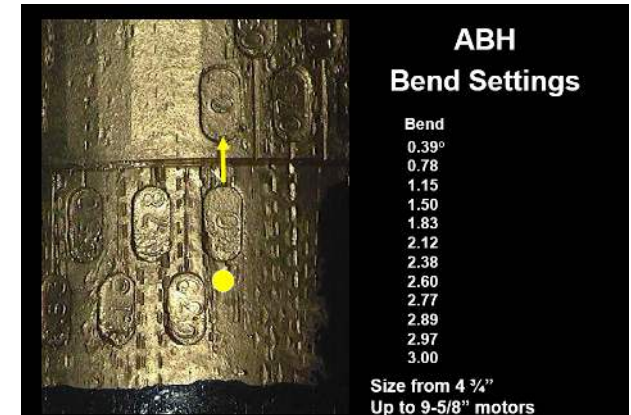
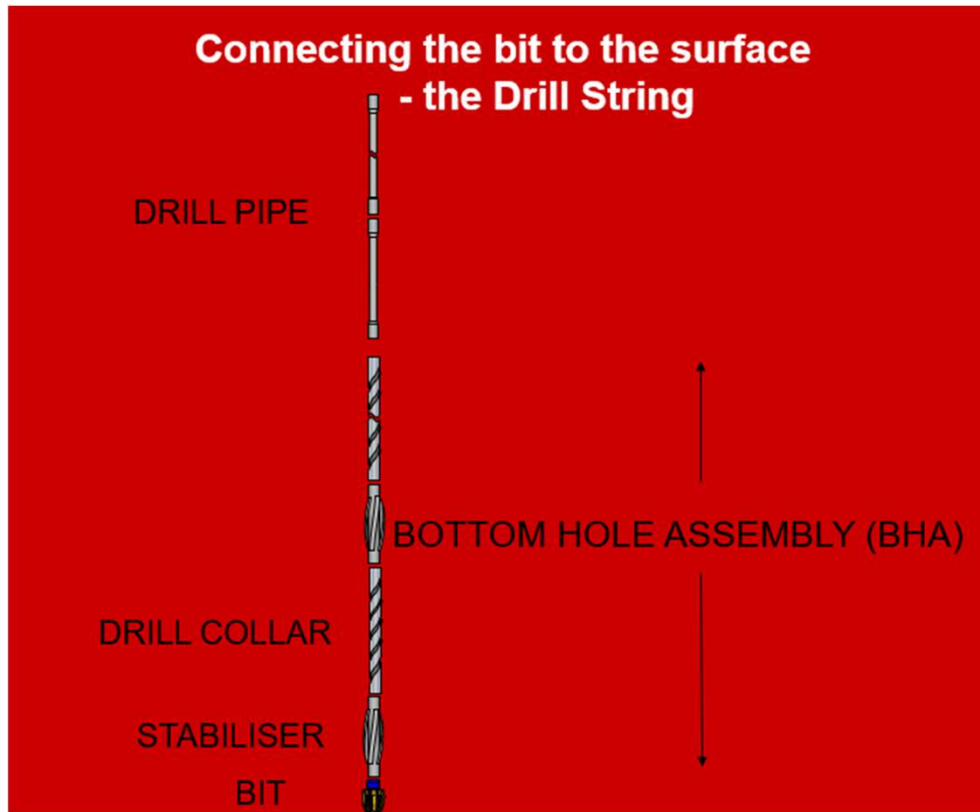


HALLIBURTON

Basic Directional Drilling

Riza Fauzi Rahman
Sperry Drilling

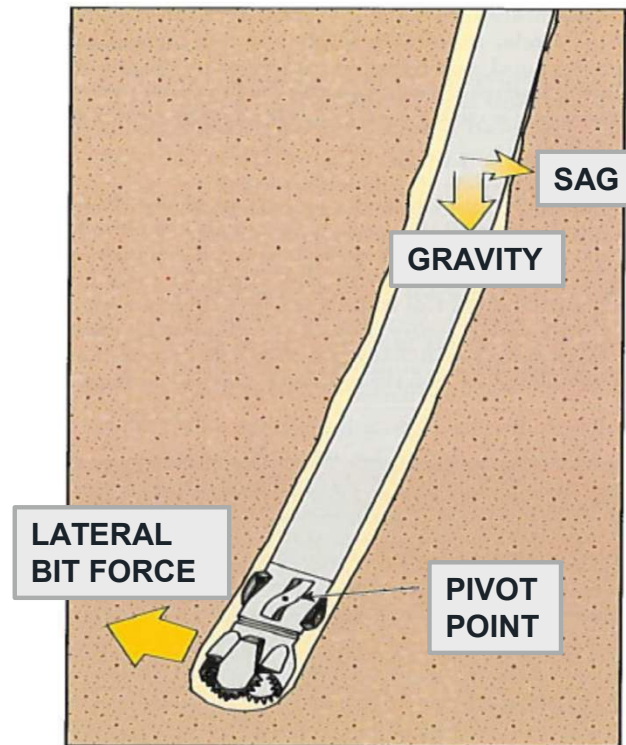
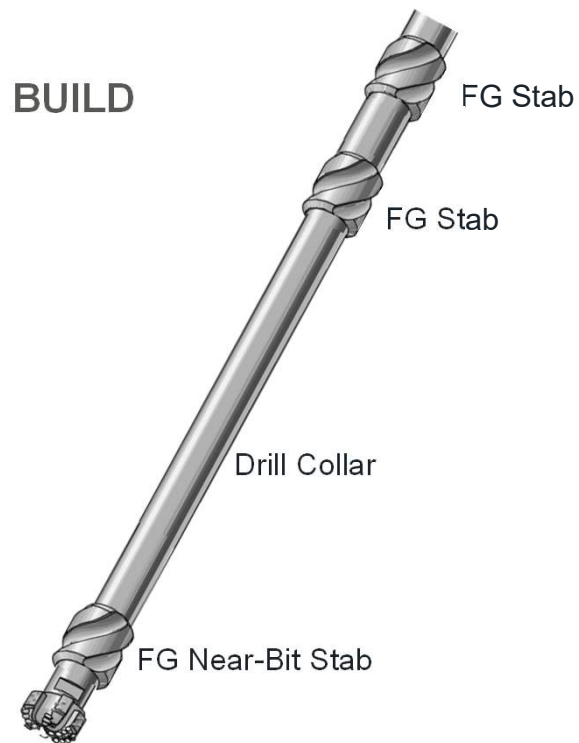
Rotary and Motor BHA



Basics of Designing a Rotary BHA

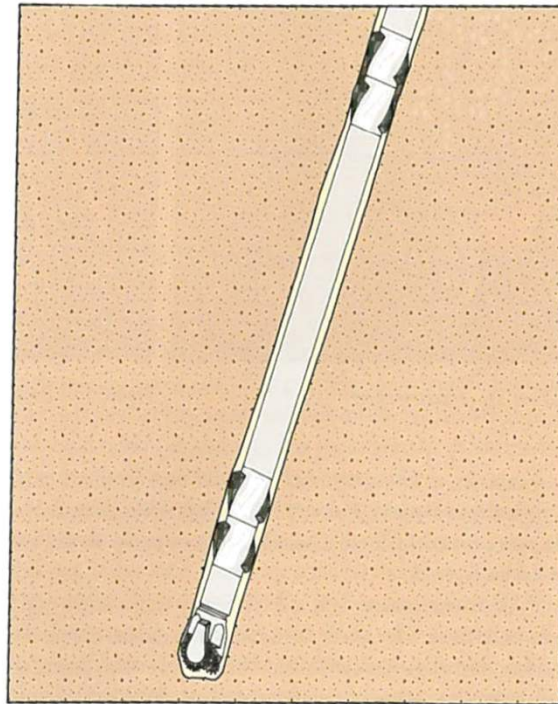
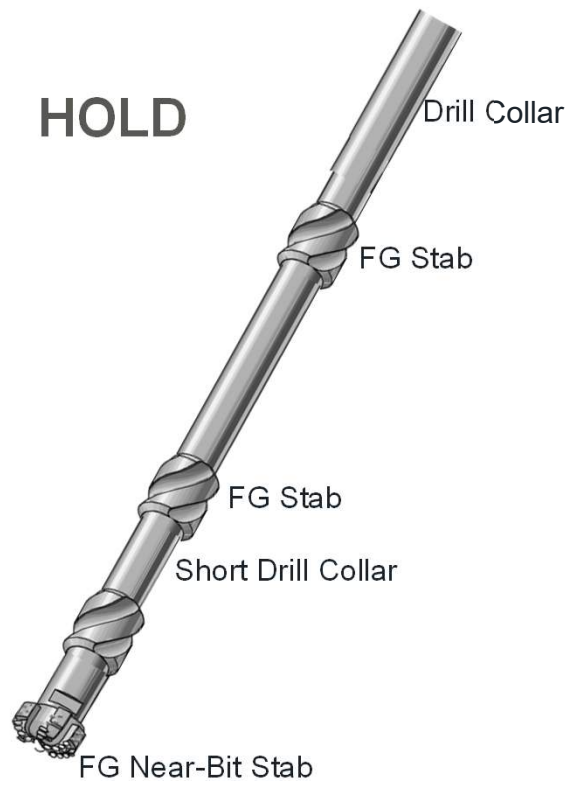
- BHA for **Building** Inclination → **Fulcrum Assembly**
- BHA for **Maintaining** Inclination → **Packed Hole Assembly**
- BHA for **Dropping** Inclination → **Pendulum Assembly**

Rotary BHA: Fulcrum Assembly



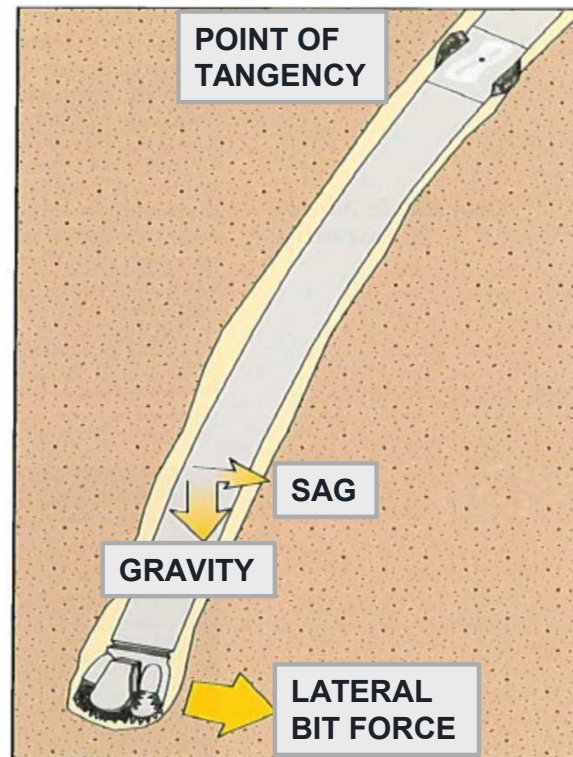
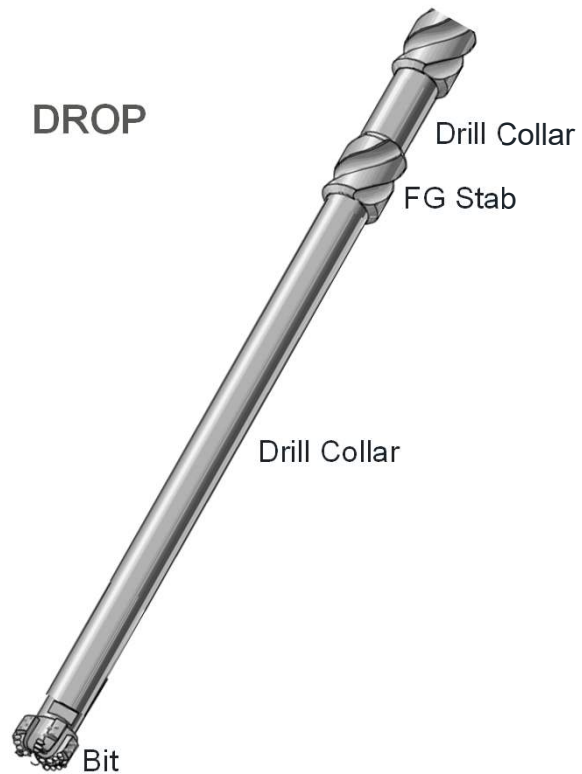
Source: Vieira, J. L. (2009). *Controlled Directional Drilling*.
Austin: The University of Texas at Austin.

Rotary BHA: Packed-Hole Assembly



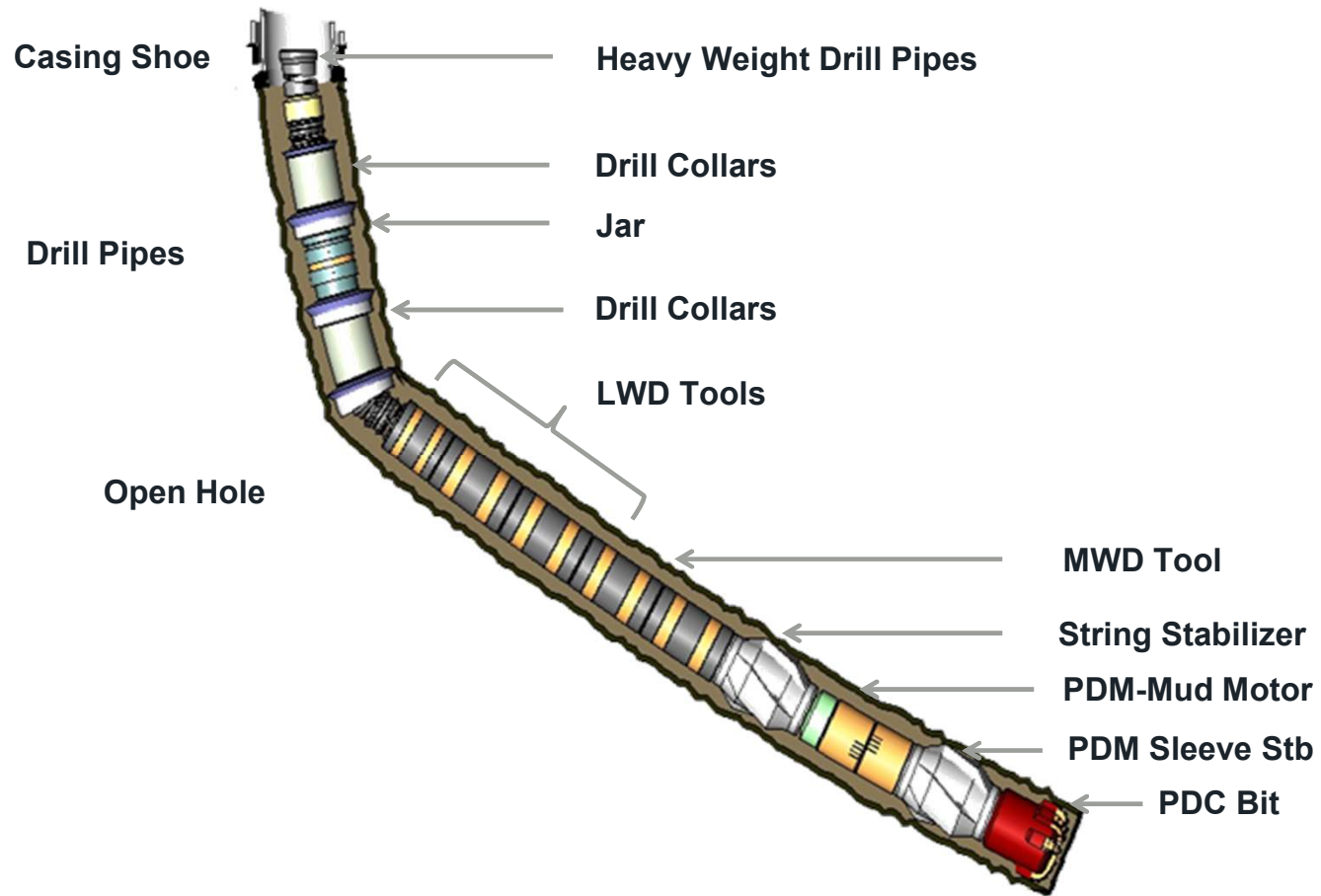
Source: Vieira, J. L.(2009). *Controlled Directional Drilling*.
Austin: The University of Texas at Austin.

Rotary BHA: Pendulum Assembly



Source: Vieira, J. L. (2009). *Controlled Directional Drilling*.
Austin: The University of Texas at Austin.

Main Components in a BHA



The Basic BHA Configurations

Dumb Iron Rotary BHA

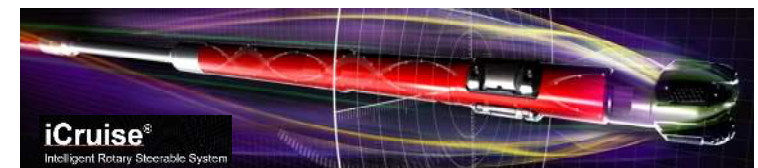
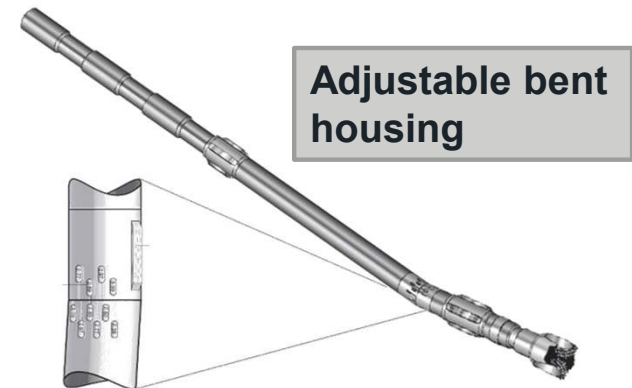
- Can be configured to build or drop within a limited range
- Little or no control over walk tendencies

Motor BHA

- Steerable
- Can generate large doglegs
- Slide drilling can be time consuming
- Limitations to the rotational speed of the drill string

Rotary Steerable Systems

- Steerable with full rotation
- Limited dogleg capability
- Susceptible to hole enlargement
- Less reliable than a motor assembly



PREPARATION - Reviewing the most updated Well Plan

- What is the importance of engaging customers and reviewing the most updated Well Plan?
 - Adhere to the plan agreed by the customer and Sperry
 - Verify the existing wells on location by matching the plan with the actual wells head (anti-collision)
 - Surface location coordinates (where the well is located on the location)
 - Referencing the correct well slot
 - Check the Datum Elevation Reference
 - RKB / Air Gap
 - Mean sea level
 - Ground level
 - Verify well trajectory, DLS, Target.

PREPARATION - - Review the Anti-Collision Process

- Importance of reviewing any anti-collision concerns related to the well plan.
 - Identify the anti-collision concerns as related to the well plan ([WM-GL-HAL-SD-AC100](#))

When $CF < 1.0$



STOP & REVIEW for HSE Risk

When $1.0 < CF < 2.0$



Ensure customer understands risks
before finalizing plans
Follow collision risk procedures

When $CF > 2.0$



Drill ahead

CF=Clearance Factor

PREPARATION - - Review the Anti-Collision Process

HALLIBURTON

Candi

Anticollision Report for P_CAND25_0003(O-O-65E)_PDSI 51.2 - P_CAND25_0003 (O-O-65E) Rev B1

Closest Approach 3D Proximity Scan on Current Survey Data (Highside Reference)

Reference Design: Candi - P_CAND25_0003(O-O-65E)_PDSI 51.2 - Original Hole - P_CAND25_0003 (O-O-65E) Rev B1

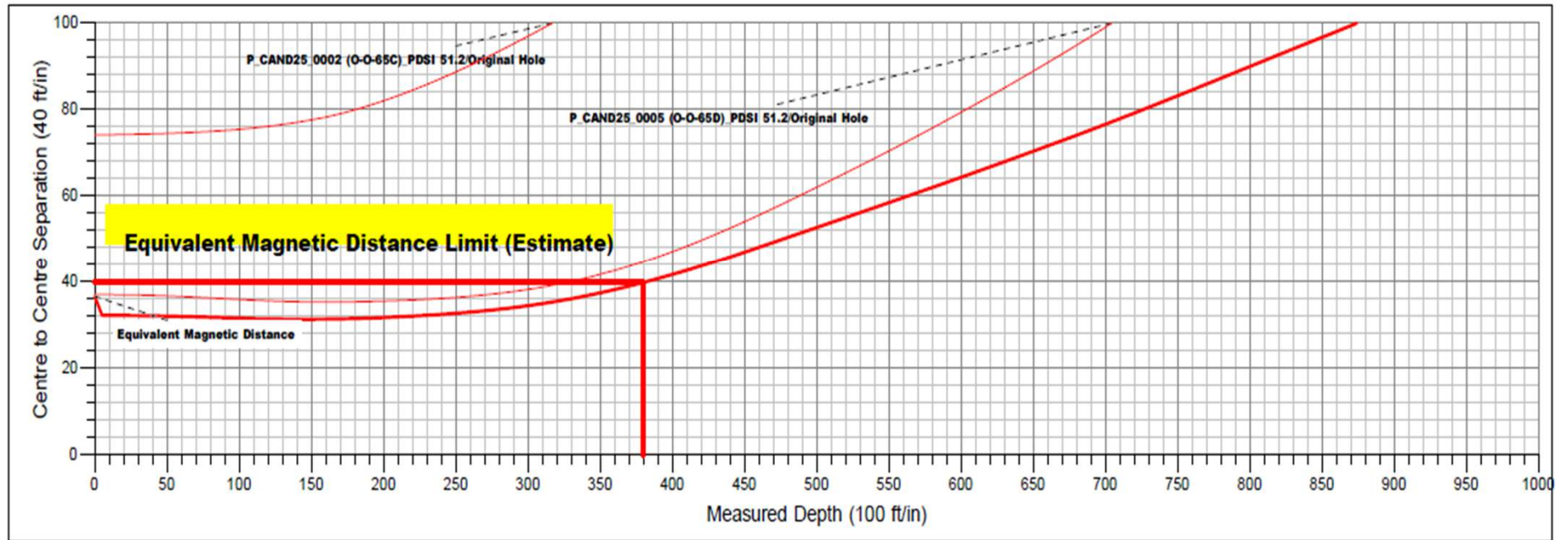
Scan Range: 0.0 to 3,834.3 ft. Measured Depth.

Scan Radius is Unlimited . Clearance Factor cutoff is 10,000.00. Max Ellipse Separation is Unlimited

Surface Error for Structure is 4.9 ft Surface Error for Well is 0.1 ft

	Reference Well			Comparison Well			Proximity			
Site Name	Measured Depth (ft)	Vertical Depth (ft)	Ellipse Major Axis/2	Measured Depth (ft)	Vertical Depth (ft)	Ellipse Major Axis/2	Between Centres (ft)	Between Ellipsoids (ft)	Relative Highside Bearing	Clearance Factor
Comparison Well Name - Wellbore Name - Design										
Candi										
CAND00013 P2 O-O-44B_ACS-010 - Original Hole -	3,834.3	3,435.1	33.3	3,280.0	2,628.2	38.9	2,339.2	2,281.3	-55.69	40.411
CAND00014 (O-O-65B) - Original Hole - Original Ho	2,620.0	2,365.1	21.4	2,798.9	2,302.9	29.4	299.9	253.6	-74.24	6.481
Candi # 10D (O-O-86A) - Candi # 10D (O-O-86A) We	3,395.0	3,048.0	29.0	1,562.4	1,552.9	9.6	3,618.5	3,595.8	143.60	159.331
Candi # 11D (O-O-65A) - Original Hole - Candi 0011(	620.0	602.3	3.8	599.9	599.9	19.8	321.2	296.4	138.03	12.948
Candi # 12D (O-O-44A) - Candi # 12D (O-O-44A) W	3,834.3	3,435.1	33.3	3,198.5	2,707.3	60.1	1,988.5	1,908.9	-59.03	24.996
Candi 001 (O-O-44) - Wellbore #1 - Wellbore #1	3,540.0	3,175.8	30.4	3,694.0	3,461.3	75.5	1,098.8	1,001.9	-140.93	11.339
Candi 002 (O-O-65) - Original Hole - Original Hole	1,315.0	1,215.2	9.1	1,221.5	1,221.5	53.1	526.5	466.4	147.18	8.761
Candi 003 (O-O-85) - Wellbore #1 - Wellbore #1	3,590.0	3,219.9	30.9	3,119.0	3,078.4	18.0	2,823.7	2,788.5	156.22	80.352
Candi 004 (S1-O-26) - Wellbore #1 - Wellbore #1	3,834.3	3,435.1	33.3	3,568.0	3,500.3	25.6	4,211.1	4,167.9	147.21	97.400
Candi 005 (S1-O-25) - Wellbore #1 - Wellbore #1	3,834.3	3,435.1	33.3	2,485.4	2,385.0	21.1	5,039.4	4,998.9	144.43	124.405
Candi 006 (O-O-86) - Wellbore #1 - Wellbore #1	3,834.3	3,435.1	33.3	3,147.2	3,017.2	29.2	3,438.6	3,388.0	141.09	67.880
Candi 007 (S1-E1-21) - Wellbore #1 - Wellbore #1	3,834.3	3,435.1	33.3	2,745.0	2,669.1	16.8	5,304.2	5,261.2	119.36	123.443
Candi 008 (S1-O-26A) - Wellbore #1 - Wellbore #1	3,834.3	3,435.1	33.3	3,646.0	3,470.6	32.8	4,637.5	4,582.7	143.30	84.708
Candi 009 (S1-O-25A) - Wellbore #1 - Wellbore #1	1,380.0	1,272.4	9.7	1,295.1	1,295.1	46.5	3,811.1	3,756.8	147.57	70.069
P_CAND25_0001 (O-O-86B) CAND00017 - Original	3,834.3	3,435.1	33.3	1,735.3	1,602.3	12.3	4,460.8	4,433.9	139.60	166.039
P_CAND25_0002 (O-O-65C)_PDSI 51.2 - Original H	550.0	538.5	3.2	534.5	529.0	2.7	155.7	148.9	-125.12	23.062
P_CAND25_0004 (O-O-65F) - Original Hole - P_CAN	480.0	473.1	2.7	474.6	471.9	2.5	45.1	39.0	100.95	7.354
P_CAND25_0005 (O-O-65D)_PDSI 51.2 - Original H	415.0	411.0	2.3	410.9	402.1	2.0	49.1	43.9	-78.34	9.505

PREPARATION - - Review the Anti-Collision Process



Ellipse of Uncertainty

► Ellipse of Uncertainty

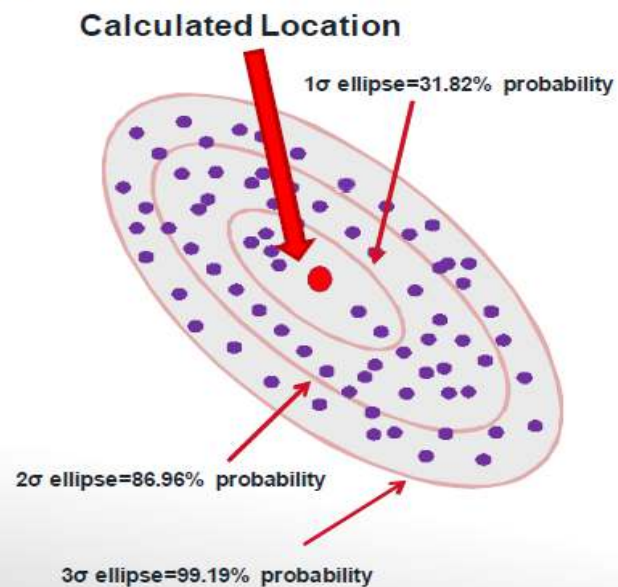
- Represents the uncertainty of the actual location of the well bore due to measurement errors
 - Surface Location Errors
 - **Instrument Performance Model (IPM)***
 - Instrument Calibration Errors
 - Instrument Misalignment Errors
 - Borehole Misalignment Errors
 - Depth Measurement Errors
 - Measuring the drill pipe at surface,
 - Stretch of the drill pipe in the hole due to its weight, and others.
 - Environmental Errors

***The MORE Survey Correction Technology used the LESS Uncertainty**

Ellipse of Uncertainty

► Ellipse of Uncertainty

- EOU can be defined as the uncertainty of the actual location of the wellbore
- 1 sigma
 - 31.82% probability in 3 dimension
- 2 sigma
 - 86.96% probability in 3 dimension
- 3 sigma
 - 99.19% probability in 3 dimension



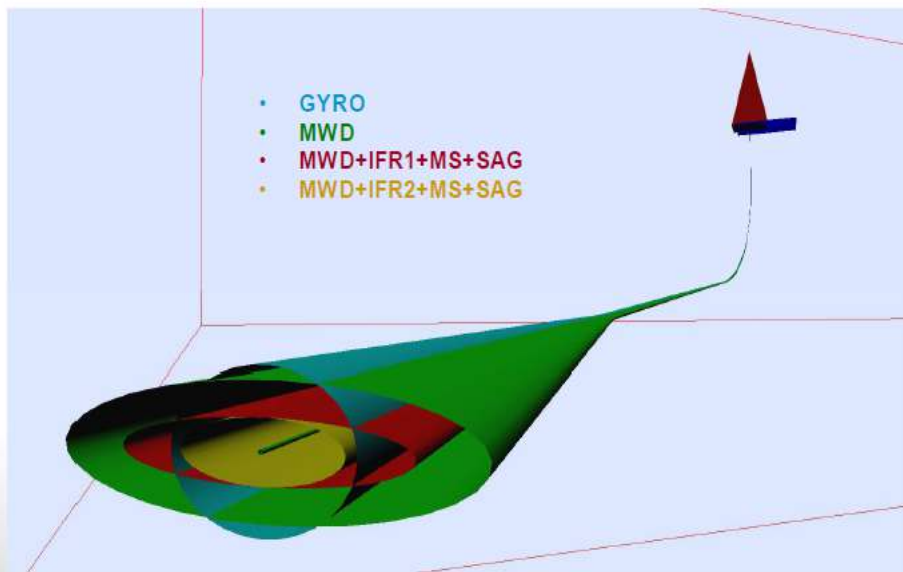
Instrument Performance Models

▶ Instrument Performance Models

- Commonly referred to as IPM, Error Model, or Toolcode
- Defines wellbore position uncertainty used in Anti-Collision Calculations
- A particular Instrument Performance Model is specific to
 - Type of survey instrument
 - Quality of the surveys taken
 - Technology applied to reduce error sources
 - Short Collar Correction (SC)
 - In-Field Referencing (IFR1)
 - Interpolated In-Field Referencing (IFR2)
 - Multi-Station Analysis (MS)
 - Sag Correction (SAG)

Instrument Performance Models

Instrument Performance Models



Using more Survey Correction Technologies decreases Uncertainty



Instrument Performance Models

▶ Common Instrument Performance Models Used

- NS-Gyro-MS: Is used for all gyro's
- MWD + SC: Is used for MWD when short collar can be applied
- MWD: Is used for MWD when short collar can not be applied

Survey Program

► Survey Program

- This is tracked by both the planned and actual wellbores
 - Planned Wellbore tracks proposed survey tools
 - Actual Wellbore tracks what was actually run
- Planning what survey tools will be run allows for more accurate Anti-Collision Scans
 - Selecting the Survey Tool selects the Instrument Performance Model (IPM) which assigns the Error Model
- Assigning the Correct Instrument Performance Model (IPM) to actual surveys is crucial to accurate Anti-Collision Scanning
- **If you are ever unsure of which Instrument Performance Model to use, contact the Survey Management Team**

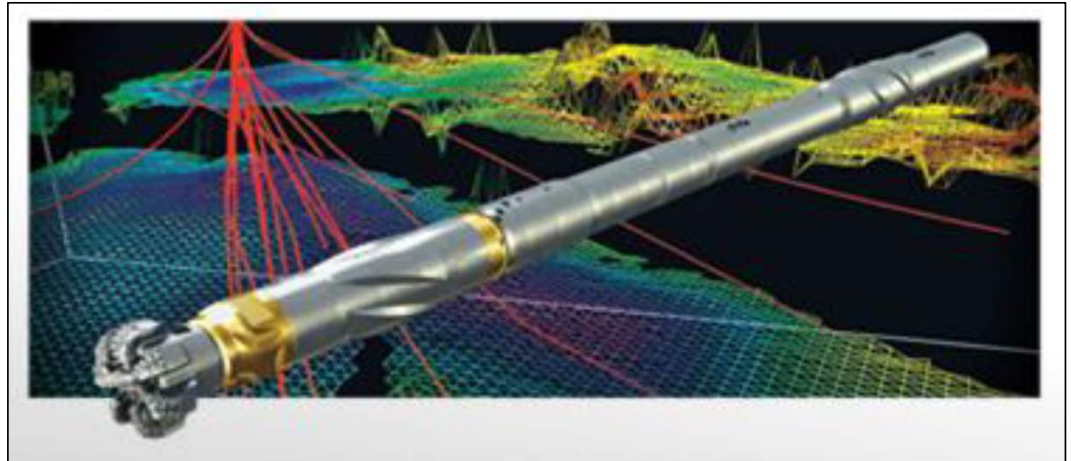
Anti-Collision

- Anti-collision is a mathematical method of avoiding a collision with an existing wellbore
- We all have the responsibility to adhere to Sperry and customers anti-collision policies and procedures
 - If a customer's anti collision policy is considered to be more stringent than this work method then an exception to this process can be requested through the Halliburton MOC process.



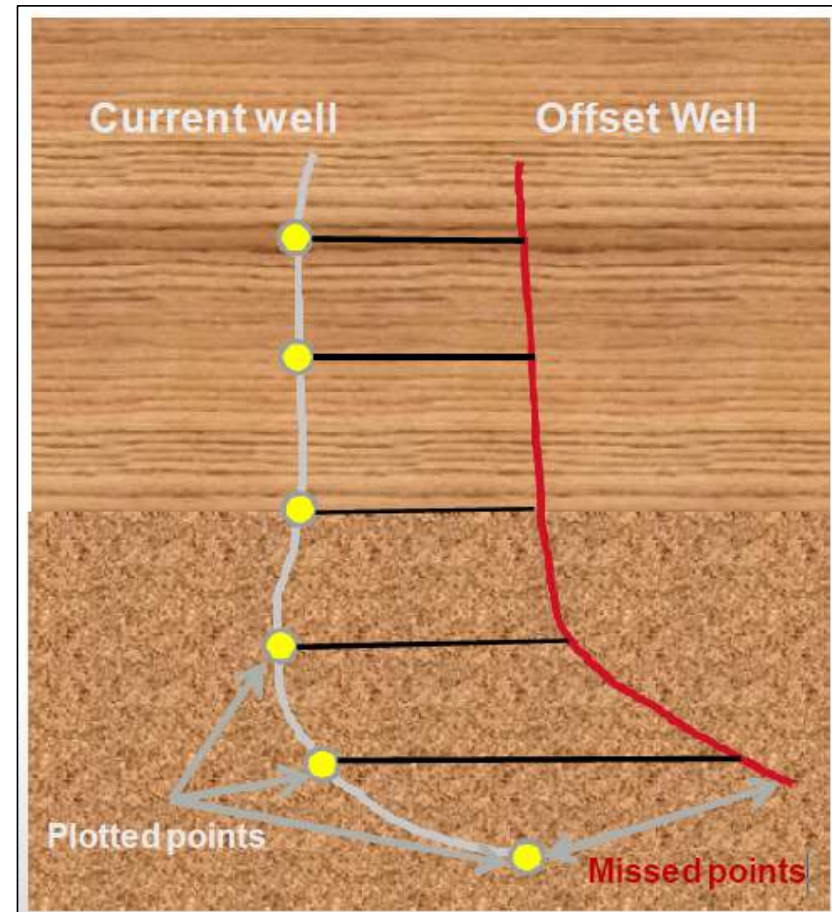
Anti-Collision Scanning Methods

- 4 Methods of scanning to avoid colliding with another well
 - Horizontal Plane
 - Travelling Cylinder & Scan
 - Closest Approach (3D)
 - Minimum Distance

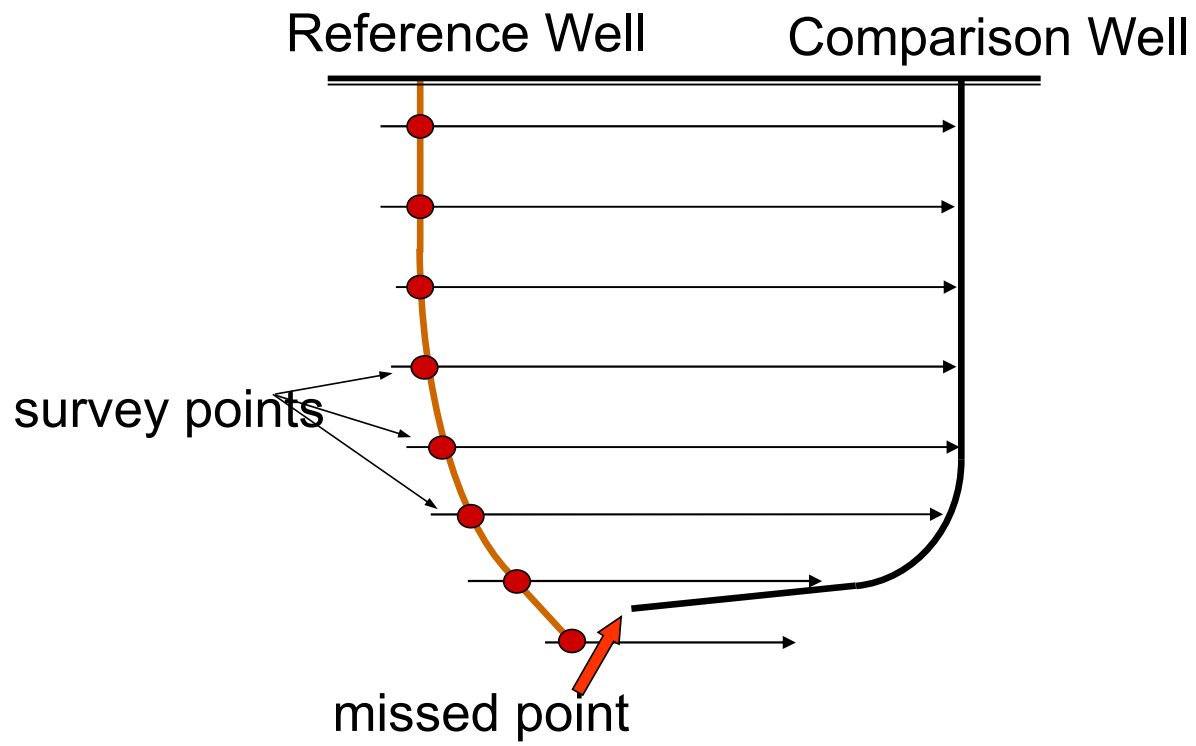


Horizontal Plane

- Horizontal Plane scanning method is intuitively the easiest to understand.
- At specified measured depth the current well is scanned down its entire length and an imaginary horizontal plane is created.
- The offset well is examined to see if it penetrates the horizontal plane.
- If it penetrates the horizontal plane then the direction and the distance where the offset well penetrates the horizontal plane to the current well is calculated.
- If the distance is less than the scan radius then the offset well needs to be plotted.



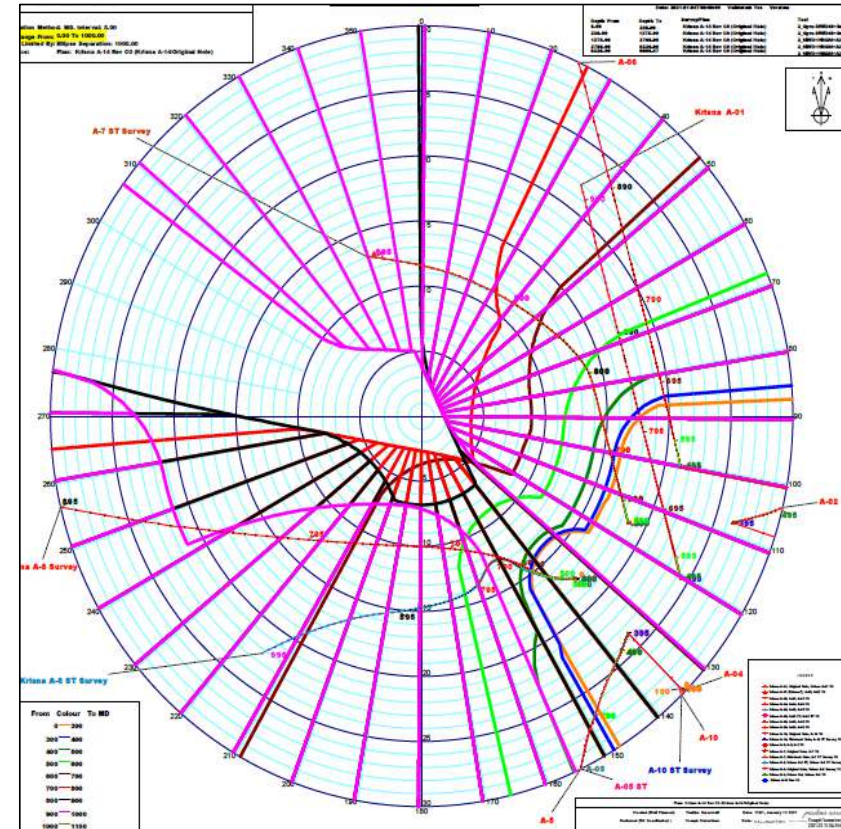
Horizontal Plane



2D

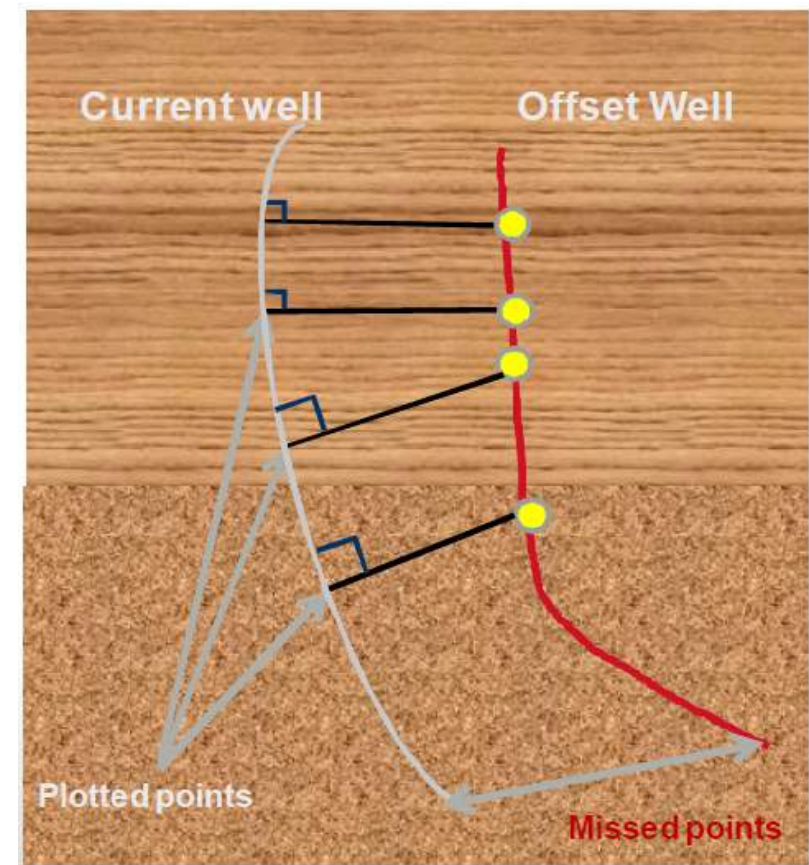
Travelling Cylinder

- The travelling cylinder is a mathematical projection used for collision avoidance
- The travelling cylinder plot show the offset wells relative to the current well
- The center is the current well
- Offset wells shown as points on the projection where they penetrate the travelling cylinder plane
- The travelling cylinder plane maybe referenced to north or high side depending on hole direction.

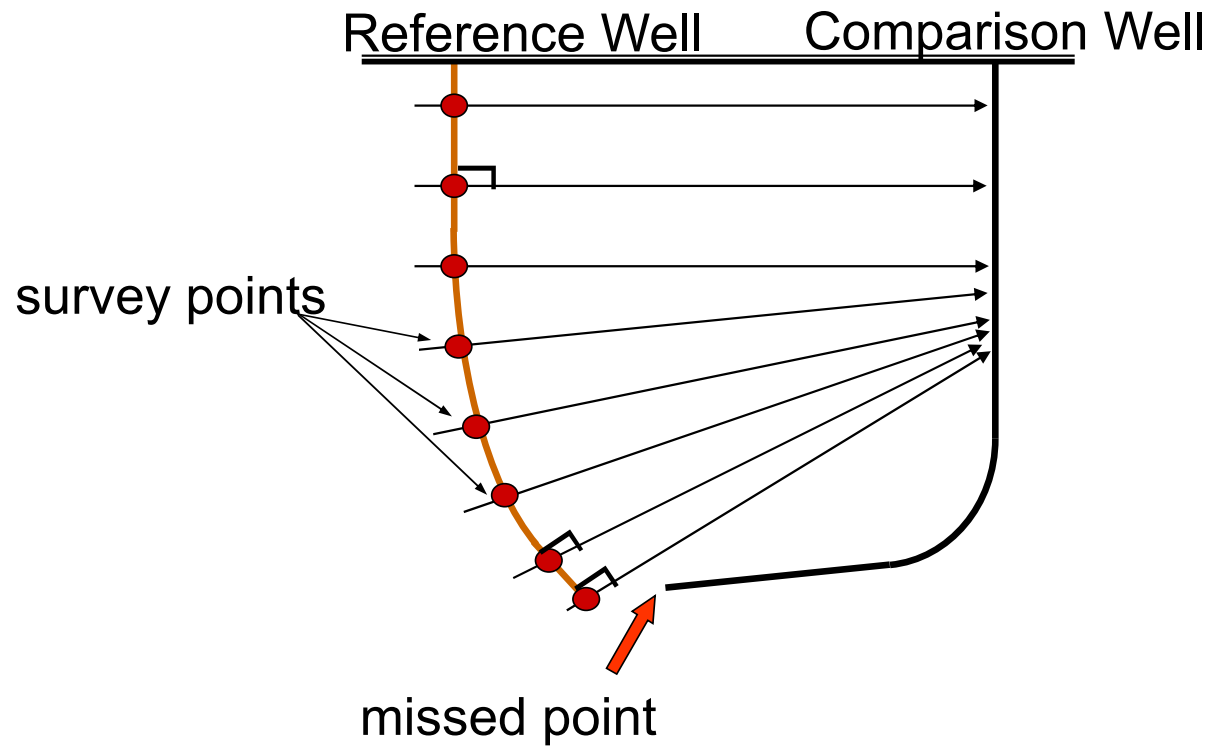


Travelling Cylinder

- At specified measured depth the current well is scanned down its entire length
- An imaginary plane that perpendicular to the current well at the measured depth is created
- The offset well is examined to see if it penetrates the perpendicular plane
- If it penetrates the perpendicular plane then the direction and the distance where the offset well penetrate the perpendicular plane to the current well is calculated



Traveling Cylinder

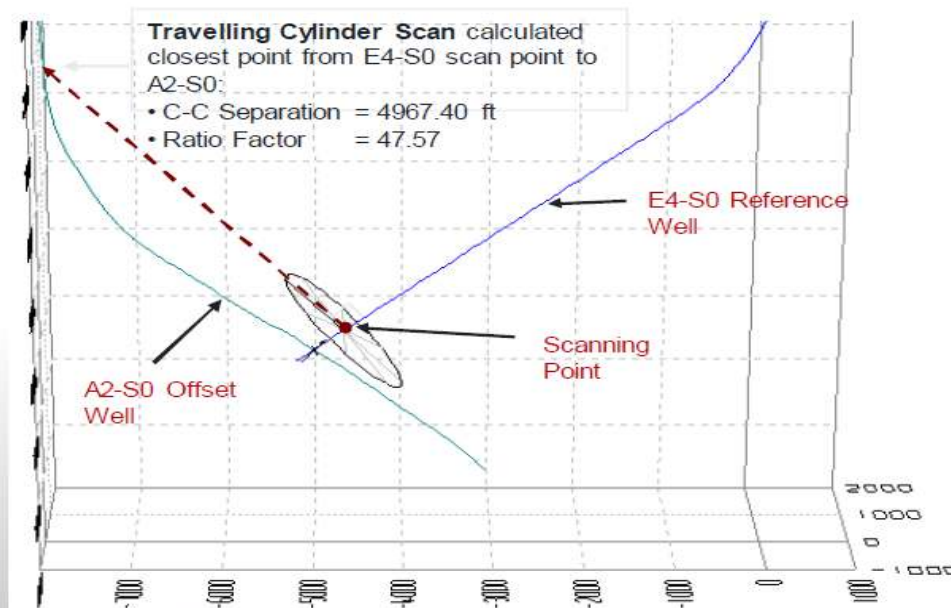


2D

Traveling Cylinder

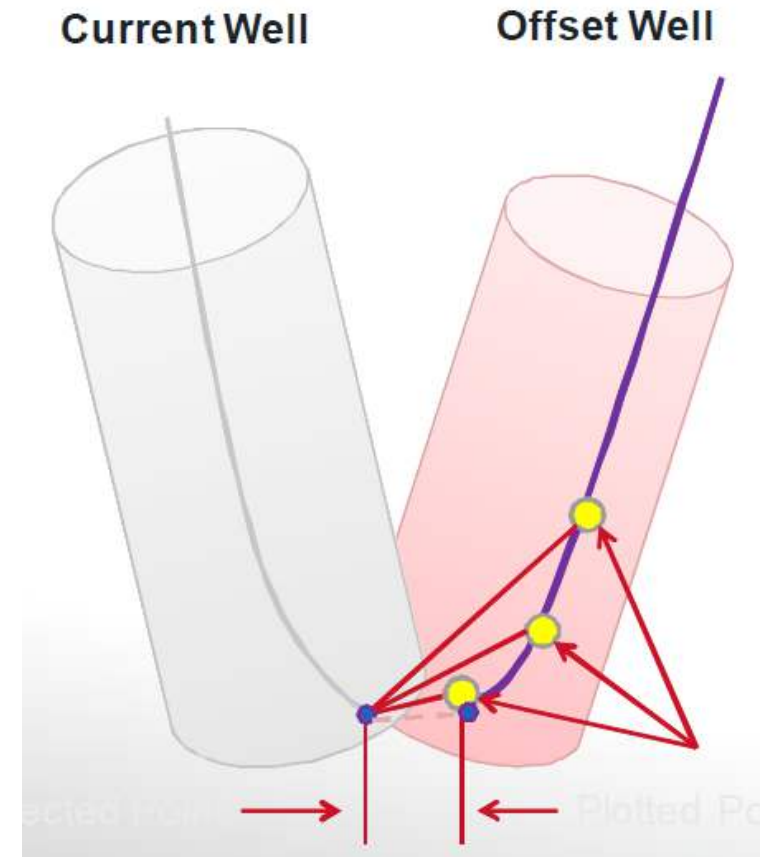
▶ Traveling Cylinder & Near-Perpendicular Intersections

- Primary deficiency with the traditional traveling cylinder method is that it can miss near-perpendicular intersections if the scan interpolation is large

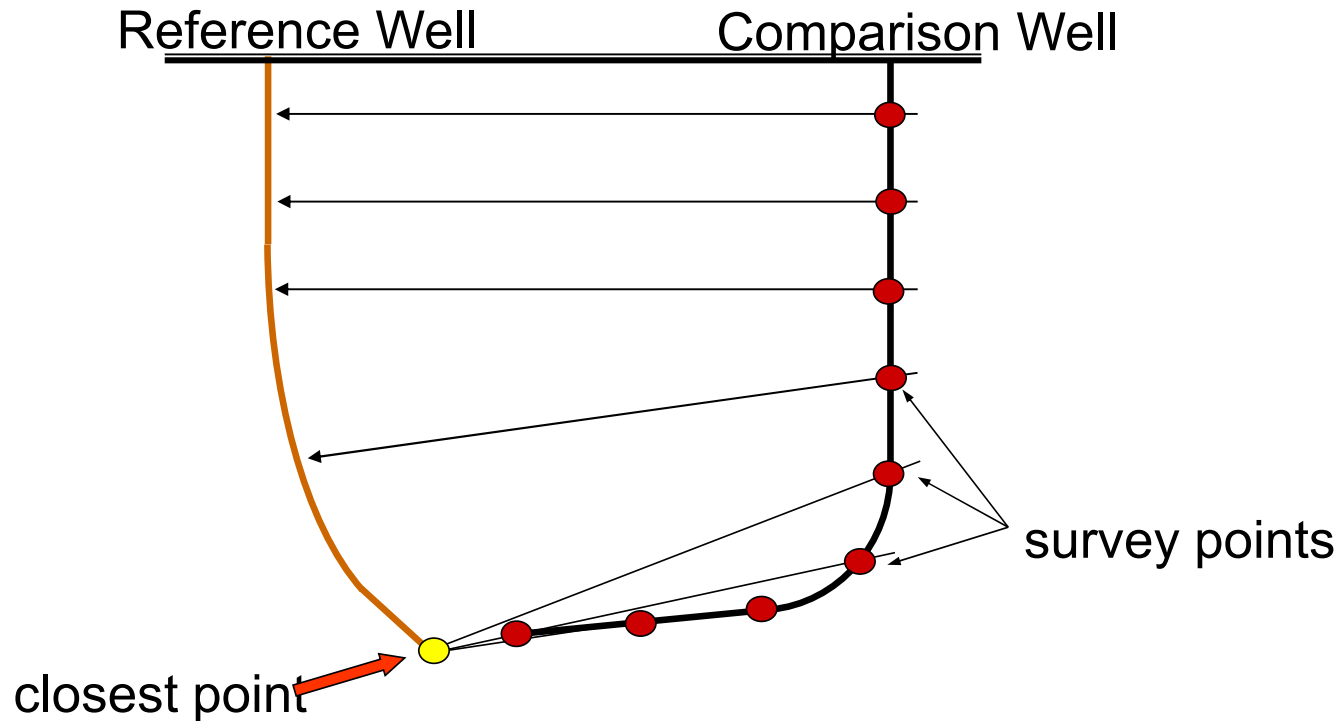


Closest Approach (3D)

- The offset well is scanned at regular measured depths down its entire length
- The current well is scanned down its entire length to find the point where its closest to the offset well
- Scanning performed in all direction in all angle to determine the closest point to offset wells.
- Considered as the safest method among others



Closest Approach (3D)



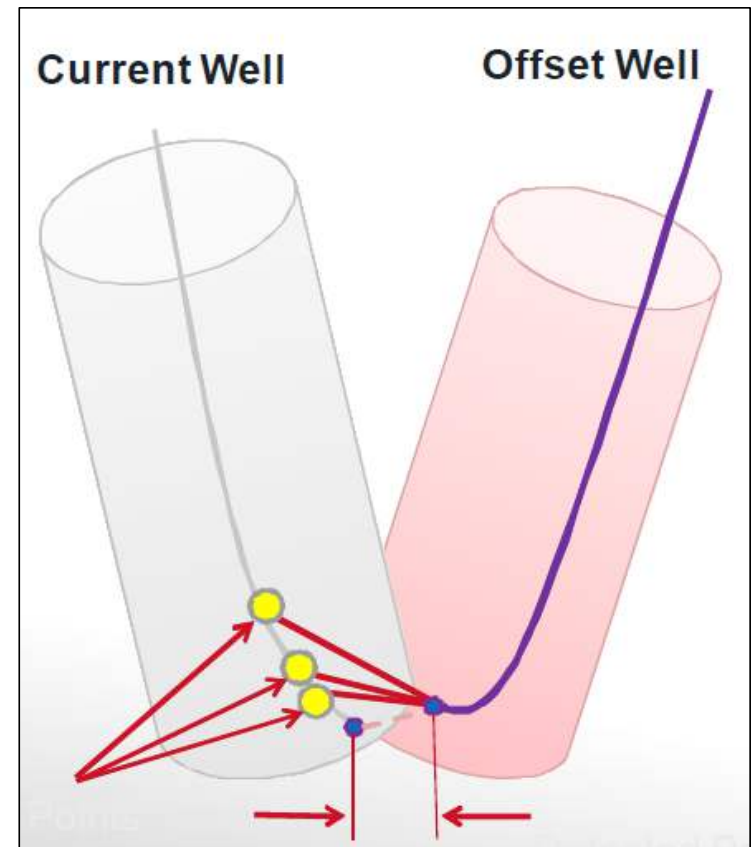
3D

Scan down from comparison well path to Reference well path

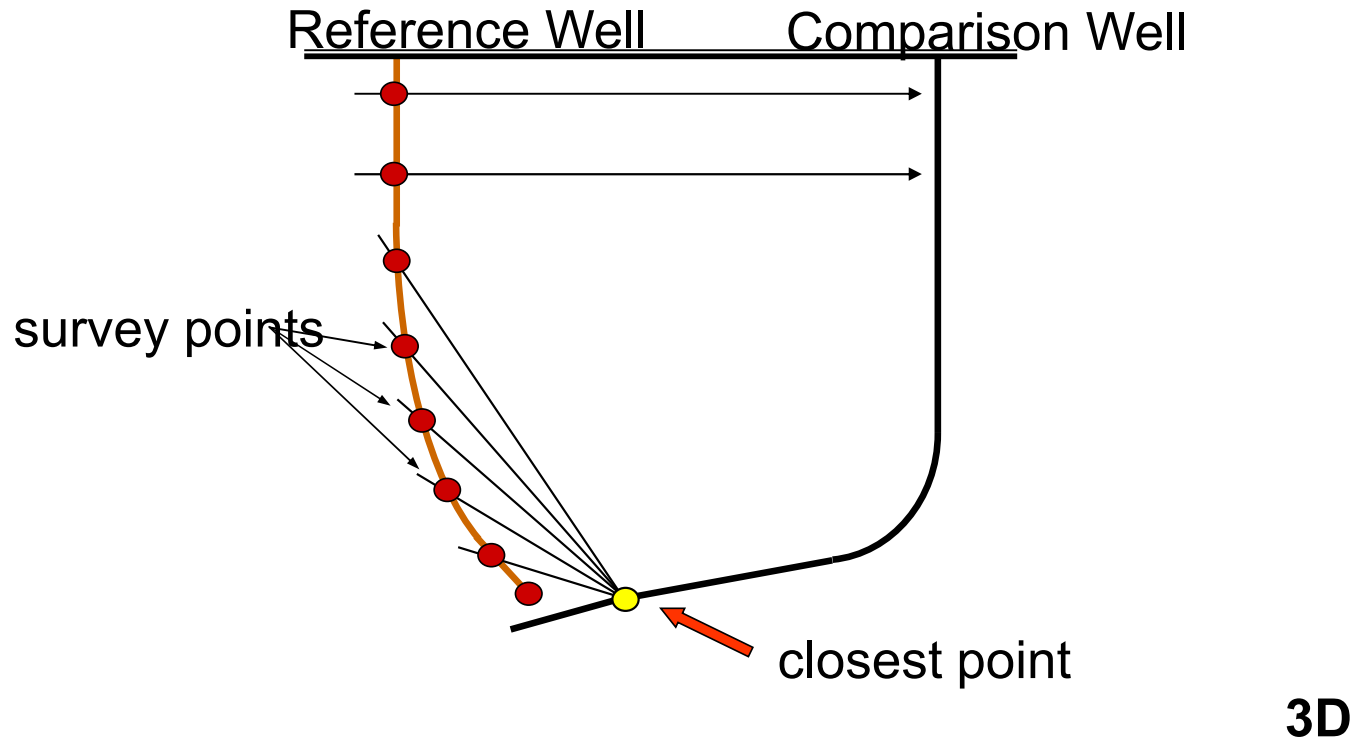
Minimum Distance

- The current well is scanned at regular measured depths down its entire length
- The offset well is scanned down its entire length to find the point where its closest to the current well
- If the distance is less than the scan radius, the point on the offset well is plotted

Not used by Sperry



Minimum Distance

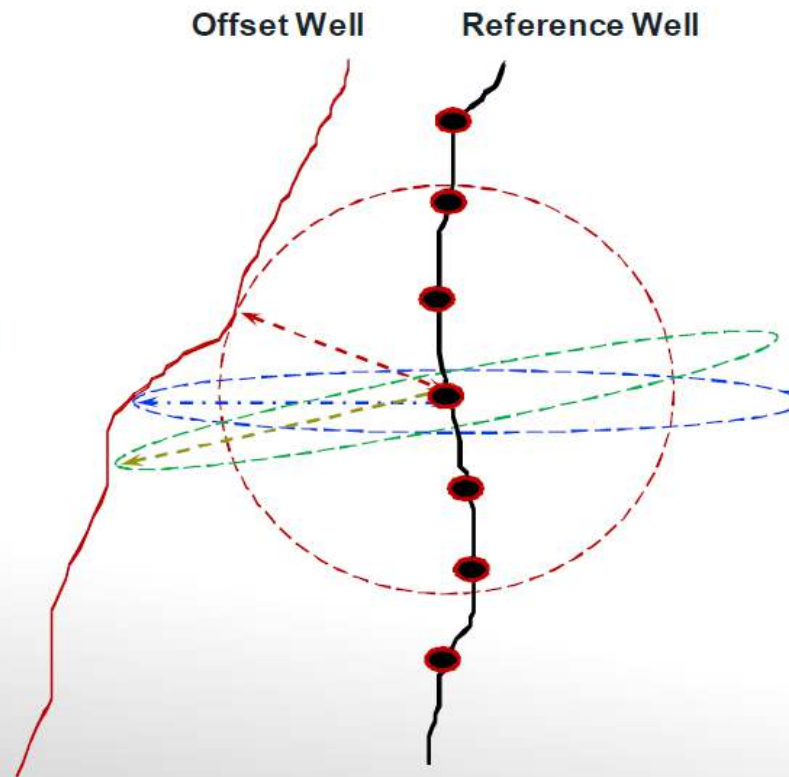


Scan down from Reference well path to Comparison well path

► Scan Method

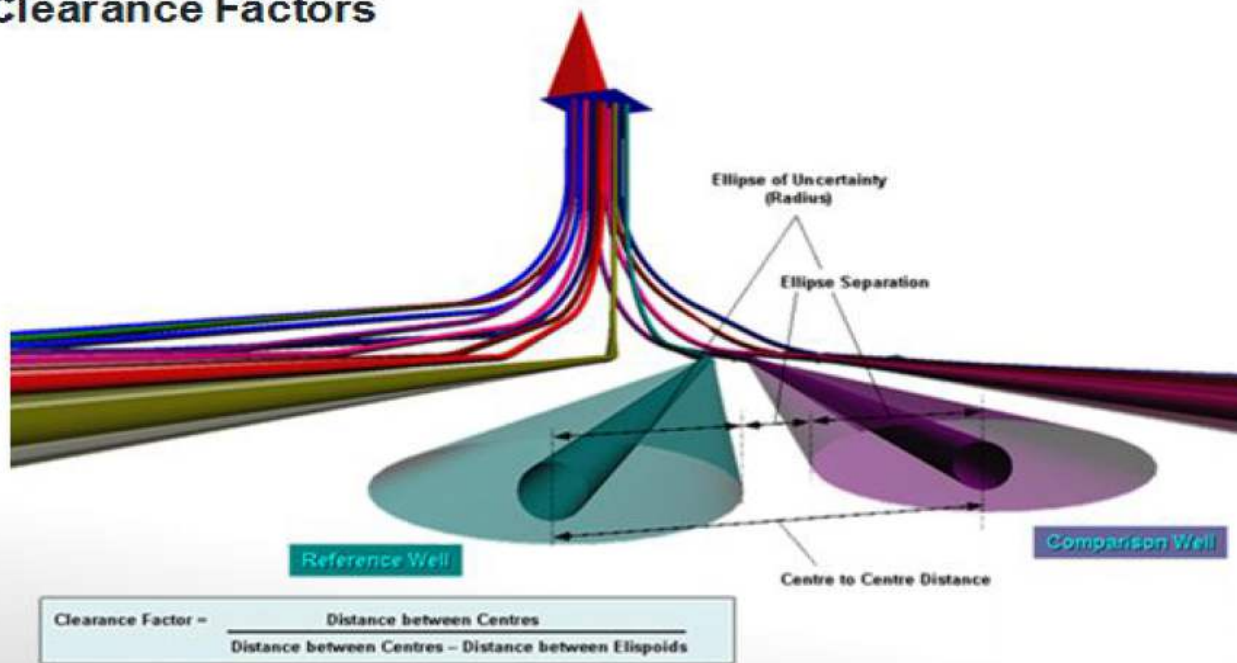
- Closest Approach 3D
- Travelling Cylinder
- Travelling Cylinder North
- Horizontal Plane

- From the same reference well scan point, the different methods have all found a different closest point, with different values of calculated wellpath separation



Clearance Factor

► Clearance Factors



- Our first criteria for determining whether, or not we have well collision considerations is the **Clearance Factor (CF)**

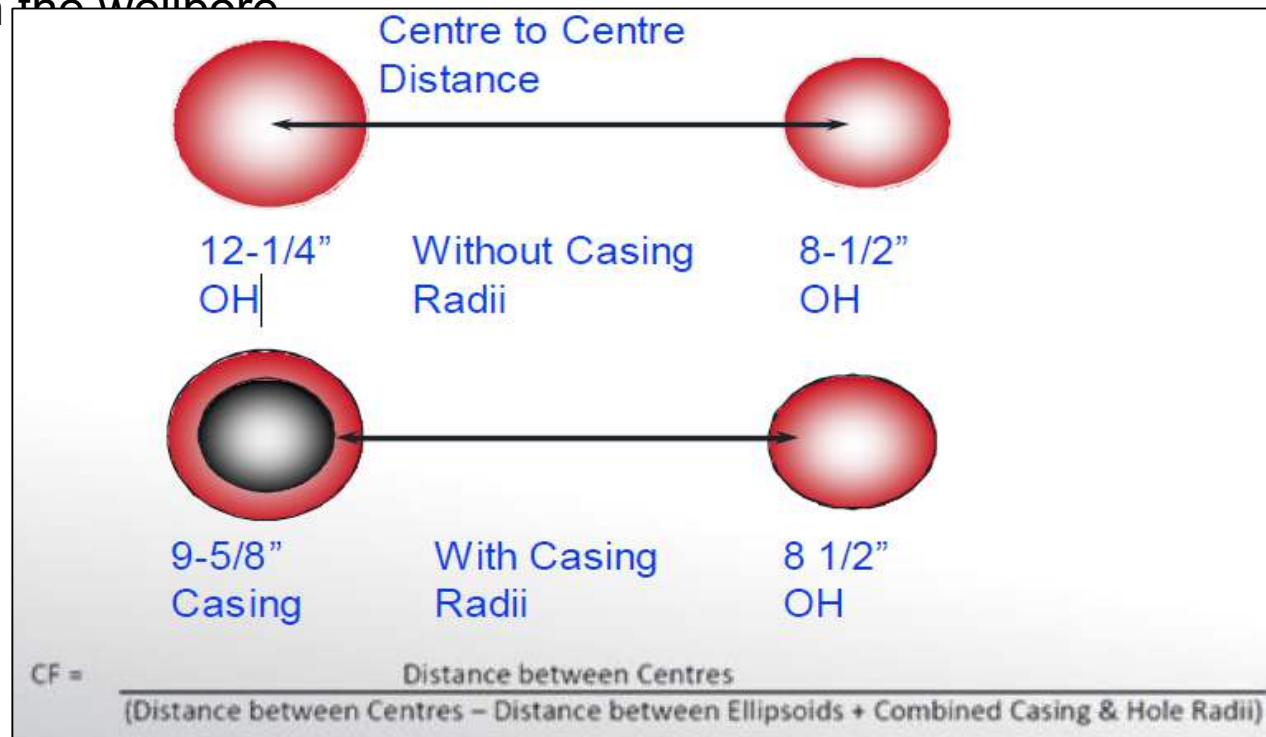
Anti- Collision Criteria

- CF ratios are used to define the separation between two wells: the current well and the adjacent
- A CF ratio is a mathematical concept that supposes that the closest distance between the surfaces of two ellipsoids of uncertainty can be determined by ordinary calculus
- The CF ratio is a dimensionless number that can be used determine the probability of a collision between two adjoining wells
- The CF must be calculated using a minimum of 2 sigma output error value and the formula for calculating the clearance factor ratio is quite simple:

$$CF = \frac{\text{Distance between Centers}}{\text{Distance between Centers} - \text{Distance between Ellipsoids}}$$

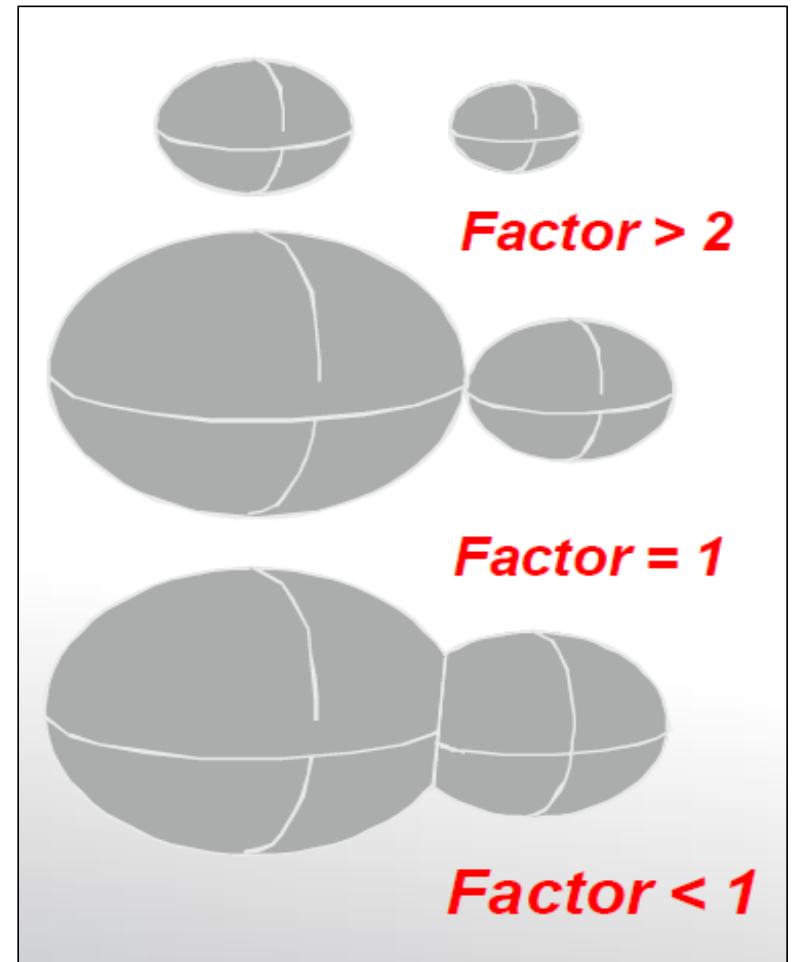
Anti- Collision

- Including casing radii in the separation factor calculation, results in the centre to centre distance being reduced by the sum of the casing radii assuming that casing is centred in the wellbore



Clearance Factor

- Our first criteria for determining whether, or not we have well collision considerations is the CF
- The smaller the CF the closer the wells are together
- CF 1.0, or smaller – wells are likely touching (1.0), or interfering with each other (<1.0)
- CF <2.0, wells are too close for comfort



Anti-Collision Properties

- **Error system** determines how the positional uncertainty is calculated
- **Error Surface** determines how the separation factor is calculated
- **Scan method** determines how the wellpath separation is calculated
- **Warning type** determines the criteria used to issue warnings (CF)

Company Properties

Partners | Wellbore Types | Audit Information

General | Anticollision | Anticollision Alerts | Calc Defaults

Survey Error Model

Error System: Output Errors are at: sigma

Input Errors are at the same confidence

ISCWSA Errors are entered at 1 sigma

Anticollision Settings

Scan Method: Error Surface:

Warning Type: Casings Type:

Warning Levels or Rules

Warning Levels are evaluated at: sigma

	Rule Name	Type	Ratio/Conf	Max Radius (m)	Def
1	No-Go Zone- Stop	Error Ratio	1.000	0.00	☐
2	Collision Avoidance	Error Ratio	1.500	0.00	☐
3	Collision Risk Proce	Error Ratio	2.000	0.00	☐
4					

OK Cancel Apply Help

Anti-Collision Safety Guideliness

- No well plan will be submitted to the customer if at any point along the well path there is a CF of less than 1.0 unless the “Permission To Drill” form has been completed by the customer and approved by Halliburton
- The well should be re-planned such that the CF is always greater than 1.0, if this cannot be achieved the customer must be informed that no drillable well can be planned until one of the following conditions are met:
 - ✓ The existing wells in the vicinity of the planned wellbore are resurveyed with a higher accuracy survey instrument and as a result no more CF’s of less than 1.0 are present along the planned well path
 - ✓ The well origin is moved to another slot such that the resultant planned well path has no more CFs less than 1.0
 - ✓ A detailed risk assessment is carried out and both the Customer and Halliburton document that they agree that the affected wells pose no safety, or environmental risk
- **Note: under no circumstances will we drill the well unless we are convinced that no safety, or environmental risk exist !!!**

Torque and Drag

Why do we use Torque and Drag?

- To evaluate the feasibility of drilling a well profile
- To determine required pipe specifications
- To determine the amount and position of HWDP
- To evaluate hole condition in real time by back calculating the Friction Factor

Torque and Drag

The Torque and Drag experienced will be a result of engineering effectiveness

- Depth (true vertical and wellbore length)
- Trajectory design
- Dimensions/lengths of tubulars set in hole
- Drill string design
- BHA components
- Actual hole shape (tortuosity)

Torque and Drag

- Friction

- What is friction?

- The force resisting the relative motion of two surfaces in contact or a surface
 - in contact with a fluid

- Two basic types of friction:

- Static friction

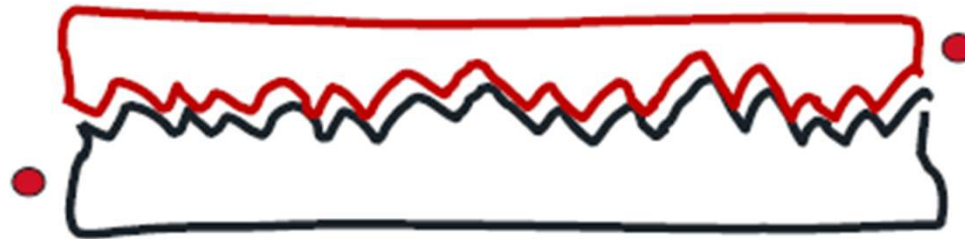
- > A force between two objects that are not moving relative to each other

- Kinetic or dynamic friction

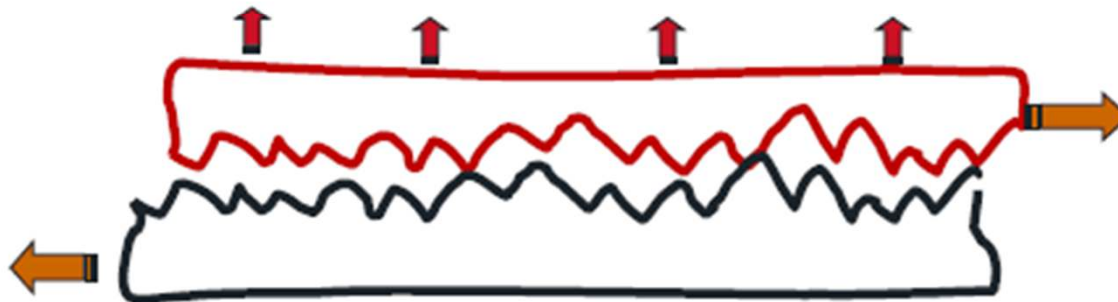
- > Occurs when two objects are moving relative to each other and rub together

Torque and Drag

Types of Friction



Static Friction



Dynamic Friction – reduced contact surface

Torque and Drag

Friction Factor

- The friction factor is a function of the smoothness or roughness of the contact surface.
- The friction factor is expressed as a value between 0 and 1, with lower numbers (0.15 - 0.22) typical for smooth surfaces (casing) and higher values (0.2 - 0.4) typical for rougher surfaces (open hole)
- The friction factor is a “lumped” parameter, meaning that it accounts for all factors that affect friction, including tortuosity (less than perfect well path), mud lubricity, polished vs. rusty casing, and hole cleaning

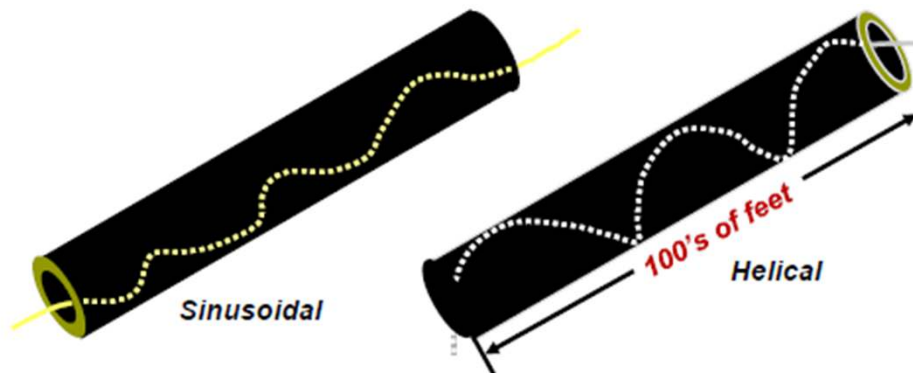
Friction Factors (Coefficient)

	Water Based Mud	Oil Based Mud	Synthetic Based Mud
Cased Hole	0.18 – 0.25	0.15 – 0.20	0.15 – 0.20
Open Hole	0.30 – 0.35	0.22 – 0.25	0.22 – 0.25

Torque and Drag

Buckling

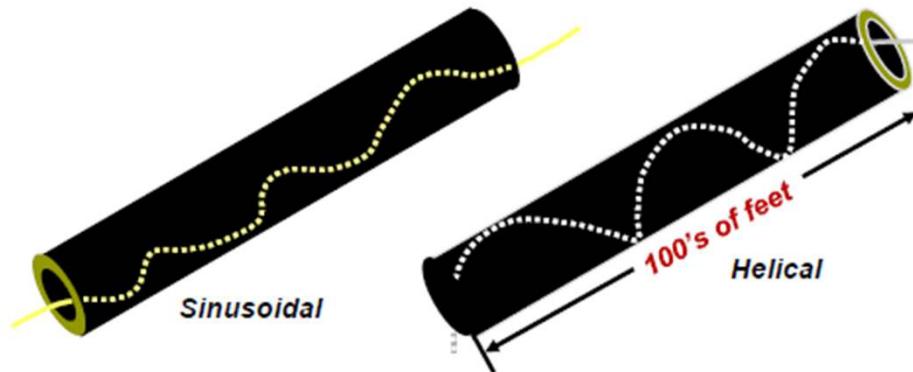
- Caused by compression and friction
- Three levels
 - 1.Sinusoidal (2-D 'snaking')
 - 2.Helical (3-D 'coil')
 - 3.Lock-up (no additional surface slackoff weight gets to the bit)



Torque and Drag

Buckling

- Caused by compression and friction
- Three levels
 - 1.Sinusoidal (2-D 'snaking')
 - 2.Helical (3-D 'coil')
 - 3.Lock-up (no additional surface slackoff weight gets to the bit)



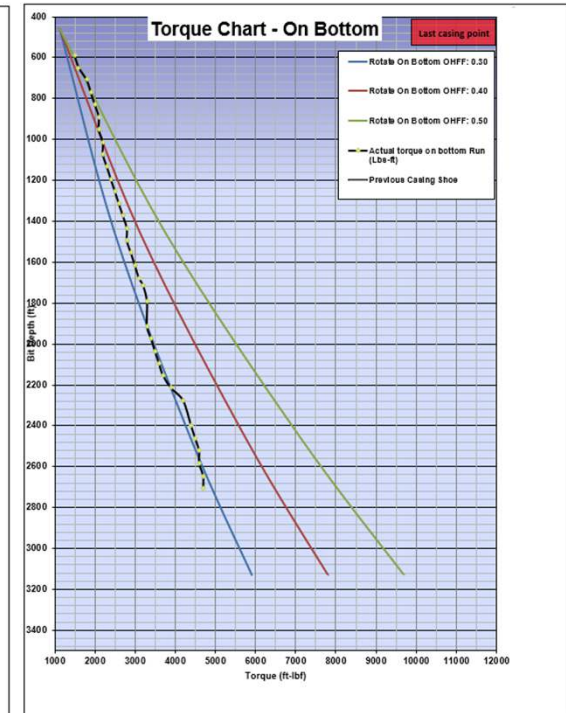
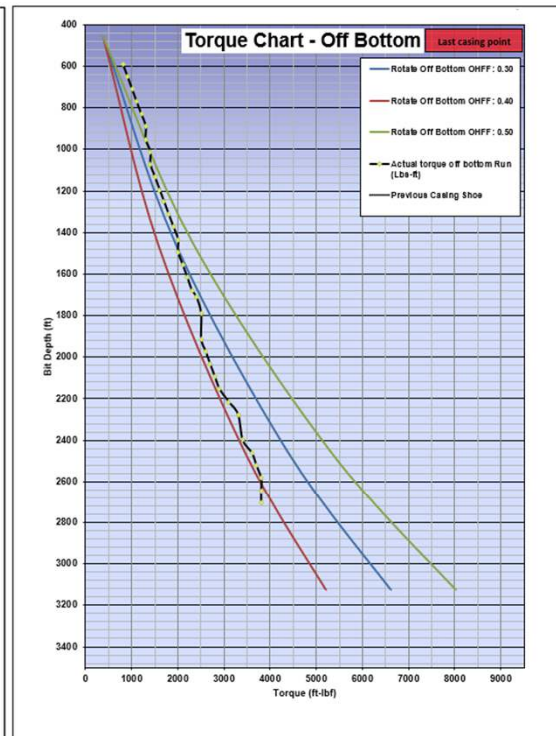
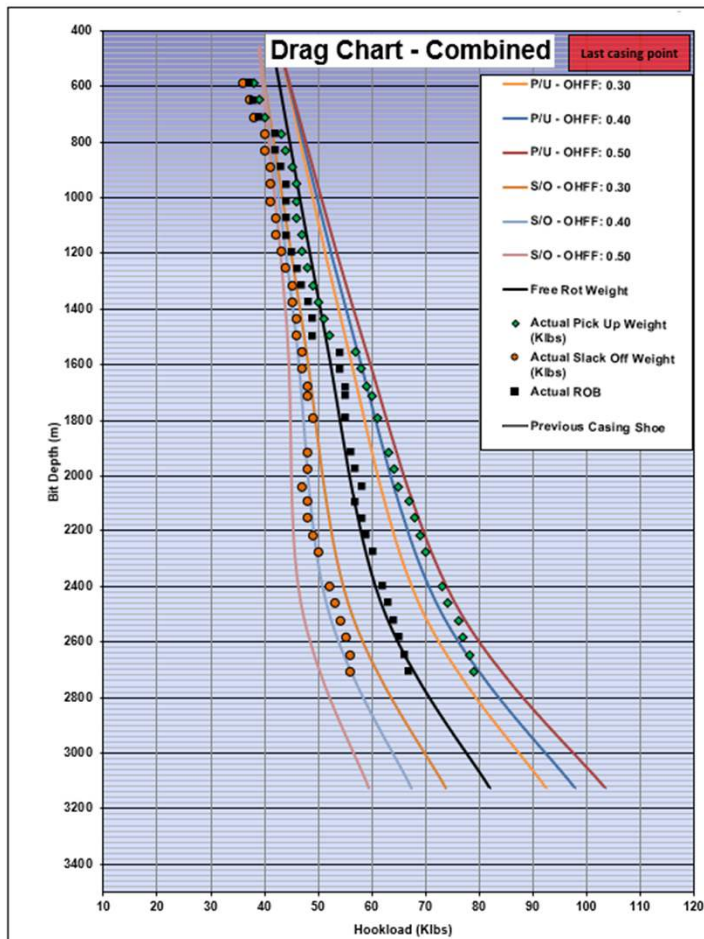
Torque and Drag

Buckling Resistance

Factors that reduce buckling :

- Increased hole inclination (gravity hold down)
- Increased stiffness (EI)
- Increased linear weight
- Decreased radial clearance (smaller hole, larger pipe)
- Reduction Wellbore friction by adding lubricants
- Improve hole quality

Torque and Drag



Hydraulics Calculations

- Wellbore Trajectory, wellbore schematic with hole size start and end depth.
- Mud properties for each hole section
- Drill string Design and limitations.
- Planned BHA and its limitations.
 - Mud motor specifications / limitations.
 - MWD specifications / limitations.
- Rig surface equipment pressure limitation.
 - pump specifications.
 - max allowable surface pressure.
- Planned Bit type and number of Jets/TFA

Input Hydraulics Simulation

Input Data:

- Select the calculations Mode
- Use the viscometer readings from the mud report.
- Calculate (n, k, Tau0, PV and YP)
- Apply the results to the hydraulics calculations.

Mud Details

Name: 8-1/2

Description:

Fluid Composition

Mud base type: Water

Base fluid: Water

Rheology Tests

Is foamable

Rheology model: Herschel Bulkley

Rheology: Rheology Fann

Import

	Temperature (°F)	Pressure (psi)	Reference
1	70.00	14.70	

Density: 9.00 ppg

Fann Data

	Speed (rpm)	Dial (")
1	600	58.00
2	300	42.00
3	200	35.00
4	100	26.00
5	6	8.00
6	3	7.00

SAMPLE TIME	Program	05:00	11:00	17:00	23:00
Depth	ft	330	656	838	1026
Mud Weight	ppg	9.1 - 9.4	9.0	9.1	9.2
Sample From :		ACTIVE	ACTIVE	ACTIVE	ACTIVE
Mud Temp	°F	90/93	96/104	105/109	109/111
Funnel Vis (sec/qt) API		40 - 55	44	51	53
Sand	%	< 0.5	0.15	0.25	0.35
Retort Water	%		95	94	94
Retort oil	%		0	0	0
Uncorrected solids	%		5	6	6
Fann 600	rpm		44	51	55
Fann 300	rpm		32	37	40
Fann 200	rpm		29	32	34
Fann 100	rpm		20	23	25
Fann 6	rpm		5	7	8
Fann 3	rpm		4	6	7
Gel 10 sec lbs/100 ft ²		8 - 10	4	7	8
Gel 10 min lbs/100 ft ²		12 - 14	6	9	10
Apparent Viscosity	cps		22	26	28
Plastic Viscosity	cps	ALAP	12	14	15
Yield Point lbs/100 ft ²		20 - 25	20	23	25
n			0.46	0.46	0.46
k			1.33	1.50	1.66

Input Hydraulics Simulation

Input Data:

- Configure the pumps
- Select the surface equipment system type from the Geometry editor.

Circulating System

Rated working pressure: 3,000.00 psi
BOP pressure rating: psi

Surface Pressure Loss - Specify: 100.00 psi

Specify: 100.00 psi
Calculate using: IADC4

	Length	
☒ Standpipe:	45.0	ft
☒ Hose:	55.0	ft
☒ Swivel:	6.0	ft
☒ Kelly:	40.0	ft
☐ Pump discharge line:	0.0	ft
☐ Top drive stackup:	0.0	ft

Mud Pit - Use average inlet temperature: 80.00 °

Return Surface Line

Mud Pumps

1. National - 9-P-100 - TRIPLEX

Pump name: National - 9-P-100 -

Volume per stroke: 2.854 gal/stk
Maximum speed: 150.00 spm
Max. discharge pressure: 3,605.00 psi
Horsepower rating: 1,000.00 hp
Volumetric efficiency: 98.00 %
Max. pump rate: 419.5 gpm

Input Data:

- Analysis Setting

Hydraulics

Pump Rates

Minimum: 400.0 gpm
Maximum: 600.0 gpm
Step size: 50.0 gpm

Pumping Constraints

☐ Use values from circulating system
☒ Use manually entered values

Max. pressure: 3,000.00 psi
Max. power: 2,880.00 hp
Max. allowable rate: 600.0 gpm

Calculation Options

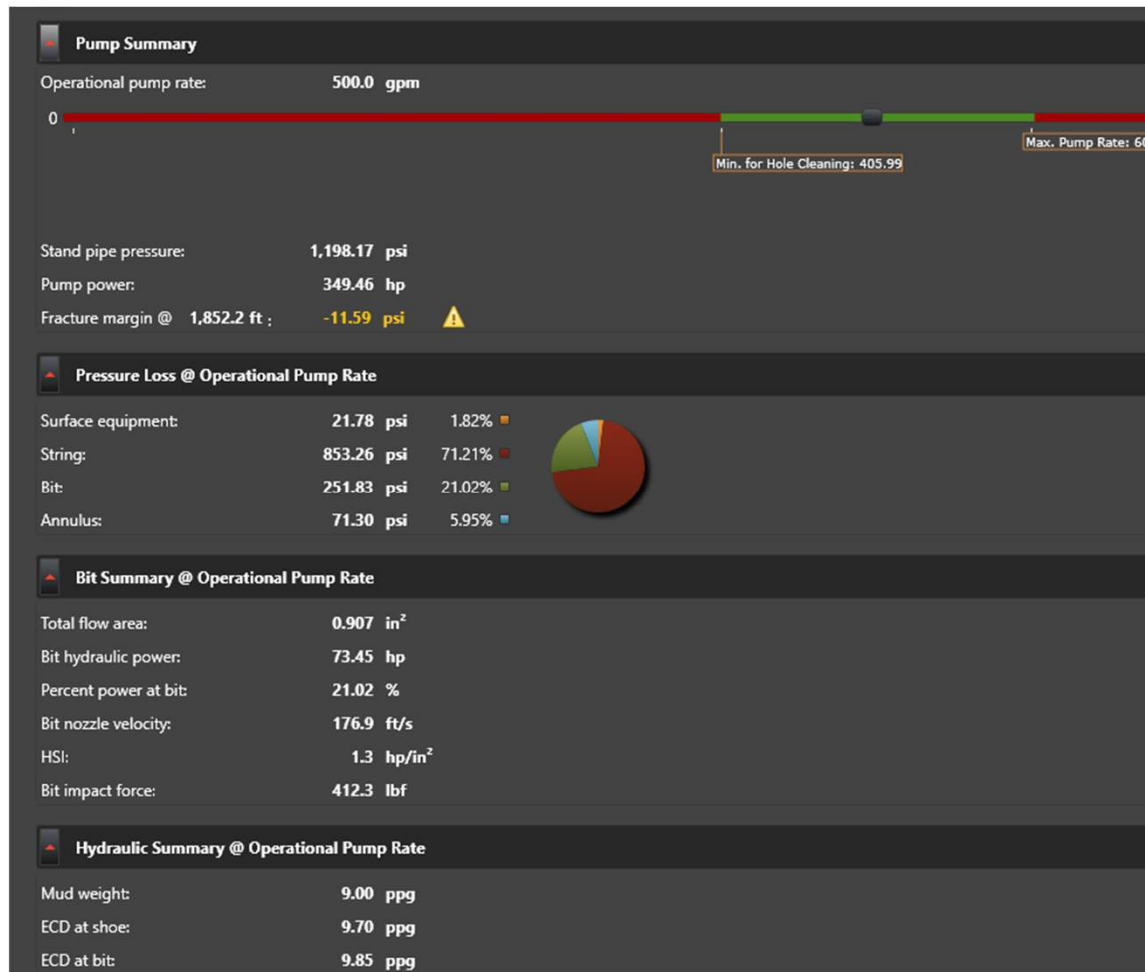
☐ Include mud temperature effects
☐ Include back pressure
☒ Include cuttings loading

Cuttings diameter: 0.250 in
Cuttings density: 2.500 sg
Bed porosity: 36.00 %
MD calculation interval: 100.0 ft
Rate of penetration: 100.0 ft/hr
Rotary speed: 90 rpm

Recommendation Flow Rate

The recommended flow rate (range) depends on the following:

1. The bit hydraulics
2. The annular velocity for better hole cleaning
3. The system pressure limit.
4. Referring to SperryDrill motor hand book (Section 2, Page 197), the recommended flow rate range will be from 850 to 900 gpm.



For a given flow rate, approximately 5 to 8% of the circulating fluid passes through the mud-lubricated type bearing assemblies for lubrication and cooling purposes. 92 to 95% of the total circulating fluid flow is available at the bit; this value for fluid flow should be used to calculate the bit pressure drop.

Bit hydraulics should be maintained with respect to the manufacturers specifications. A general guideline for rock bits is to maintain the jet velocity at around 300 feet per second. For PDC bits the horsepower per square inch (HSL) should be in the range 2.5 to 7. A rule of thumb for annular fluid velocity is 100 feet per minute with the flow being laminar over the drillpipe; a higher flow rate to aid cooling and cleaning is preferable to a high HSL.

SperryDrill and GeoForce motors are supplied with high pressure flow restrictors as standard. Low pressure flow restrictors are used for special applications.

Running low bit pressure drops (less than 200 psi) with 'high pressure' flow restrictors does not allow enough fluid to pass across mud-lubricated bearings and can result in over-heating and wear of the bearings (high pressure flow restrictors are the standard motor configuration).

Running high bit pressure drops (greater than 500 psi) with 'low pressure' flow restrictors allows too much fluid to pass across mud-lubricated bearings and can wash out the bearings (this can occur due to bit plugging).

How to Calibrate Hydraulics Simulation (Standpipe Pressure – SPP)

1. Validate All Input Data (Critical First Step)

- Before doing any calibration, make sure your inputs are correct:
 - ◆ **Mud properties (use latest lab data)** : Mud Weight (MW), Plastic Viscosity (PV), Yield Point (YP), Gel strength

2. Drillstring / BHA

- OD / ID of pipes, Length of each section, Bit nozzle sizes / Total Flow Area (TFA)

3 Well geometry

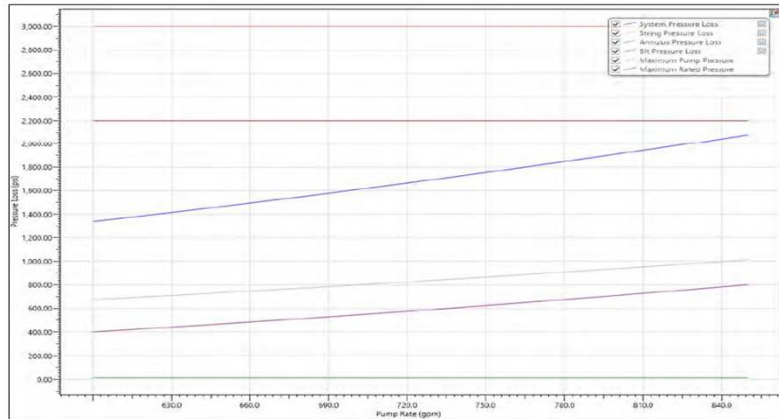
- Actual hole size (not planned), Casing ID, Open hole condition (washout?)

4. Flow rate

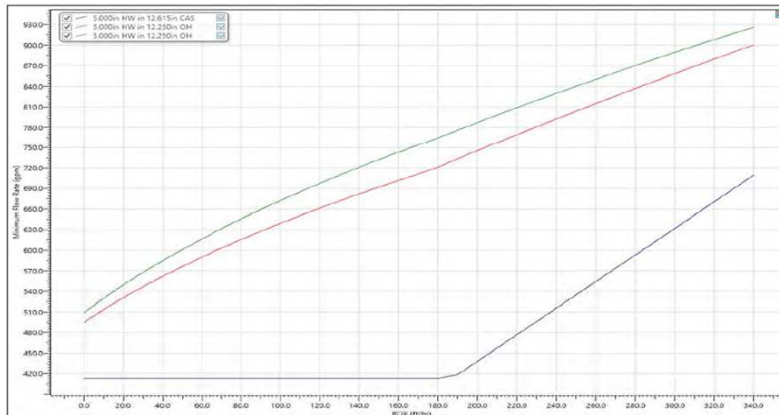
- Confirm actual pump rate (gpm or lpm)

Hydraulics Simulation

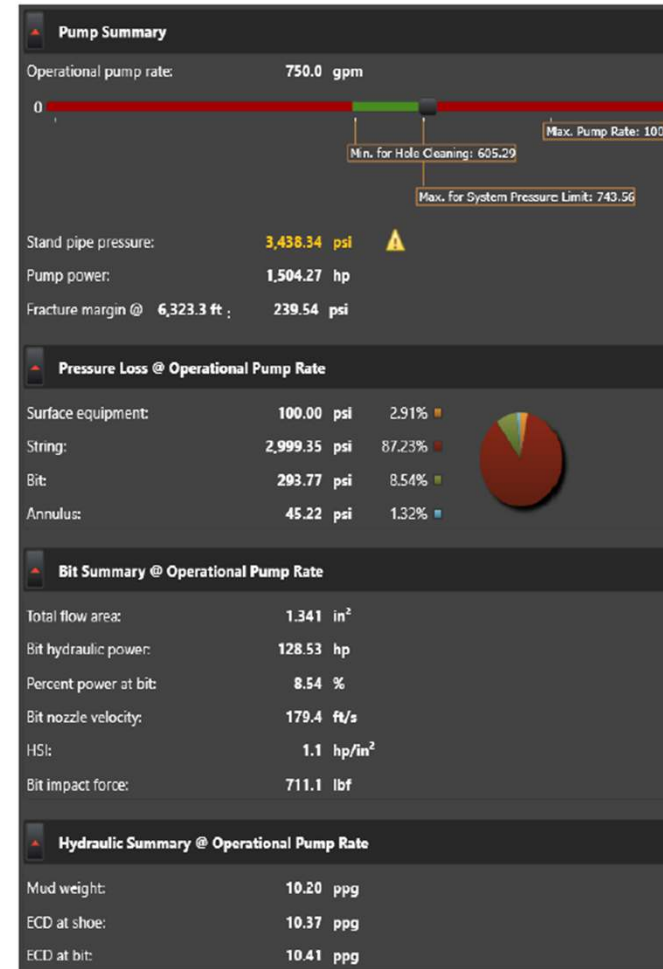
1. Pressure Loss vs Pump Rate



6. Minimum Flow Rate vs ROP

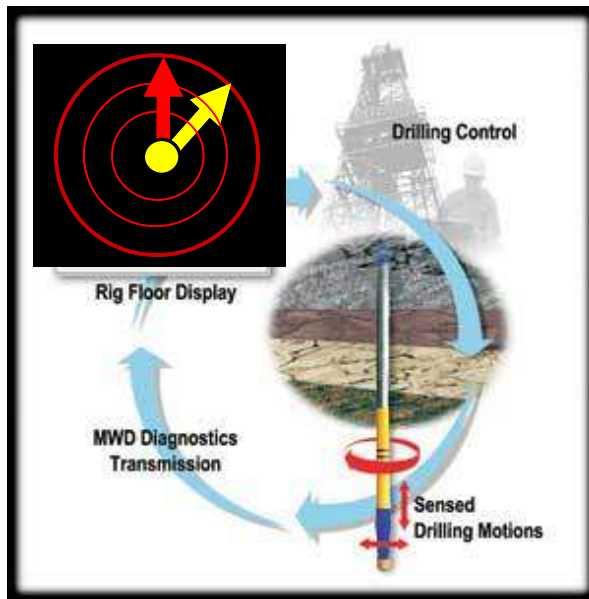


© 2018 H. Note: Minimum flow rate to clean out around 639 GPM at ROP ± 100 ft/h and MW 9.1 ppg



EXECUTION - Steering the Well

Orienting the Steerable Motors

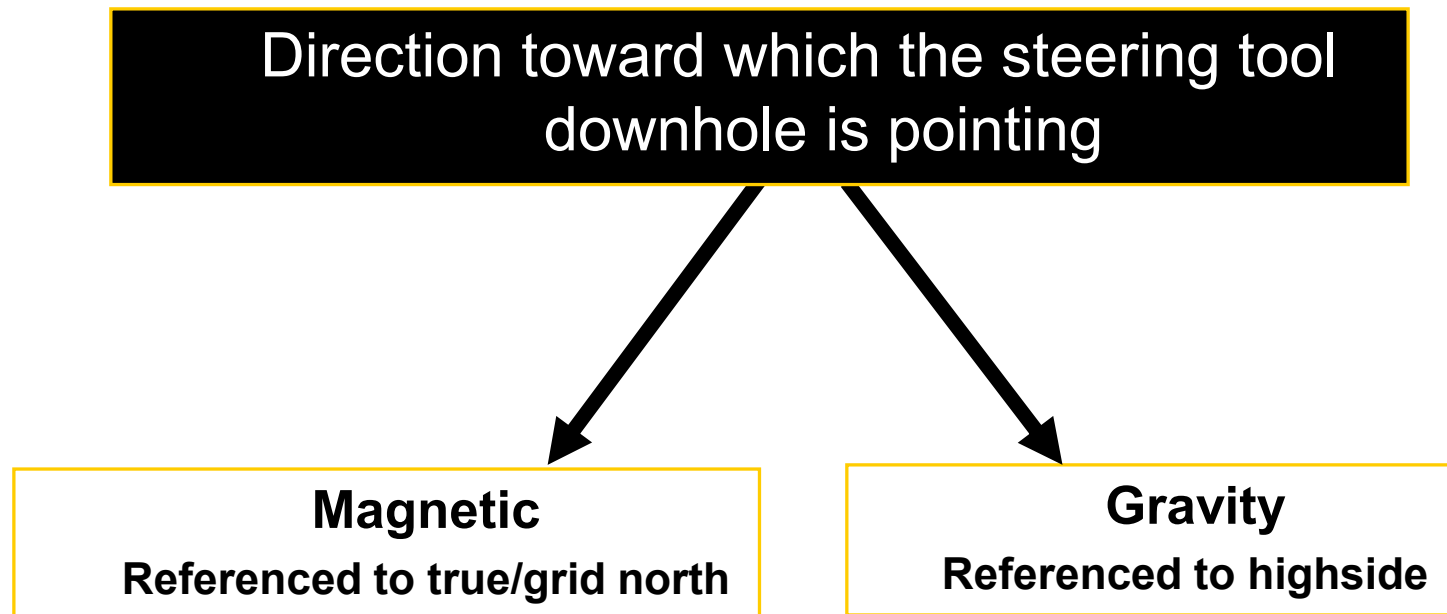


- MWD will send the RealTime toolface of the steerable motor to the surface
- The drillstring is re-oriented to the desired toolface
- Drillstring does not rotate while the bit is driven by the mud motor to drill the formation



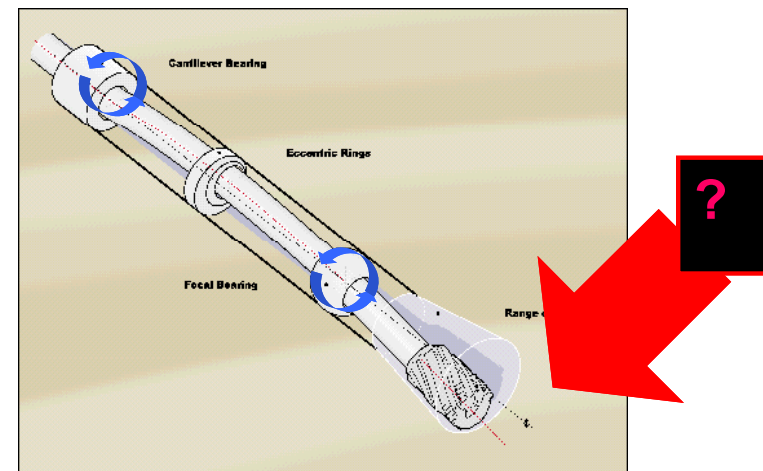
Varying the amount of sliding
And rotating interval

EXECUTION - Steering the Well

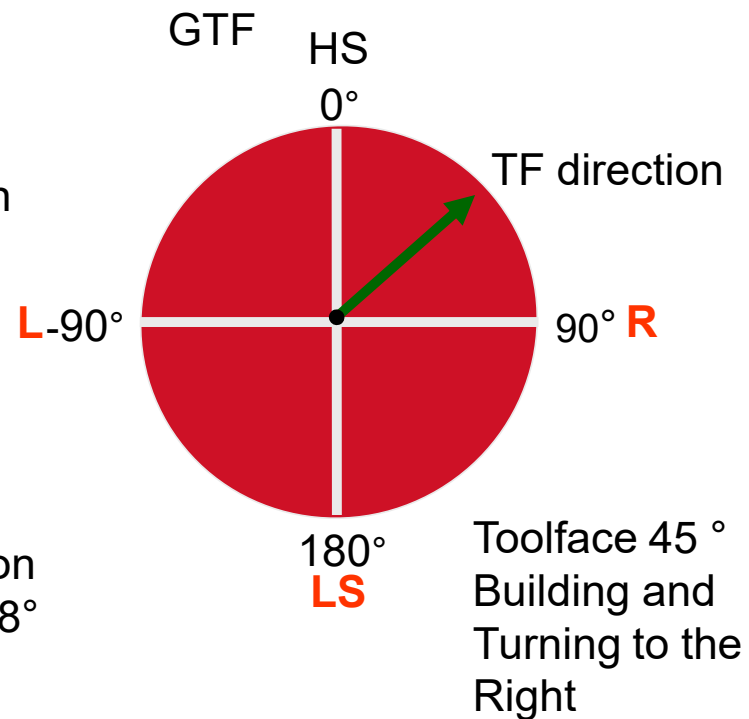
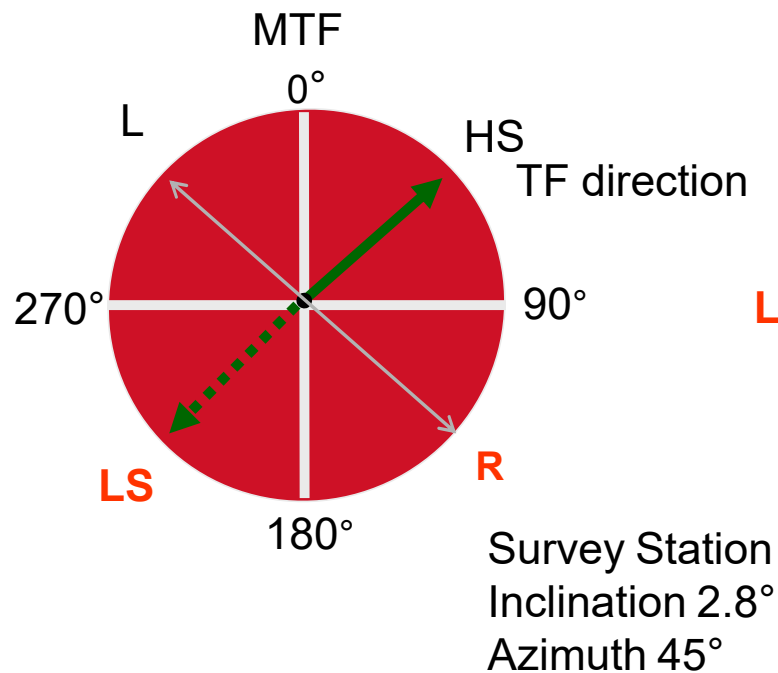


Toolface

Direction toward which the steering tool downhole is pointing

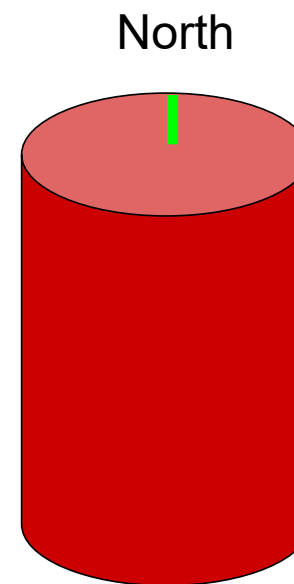
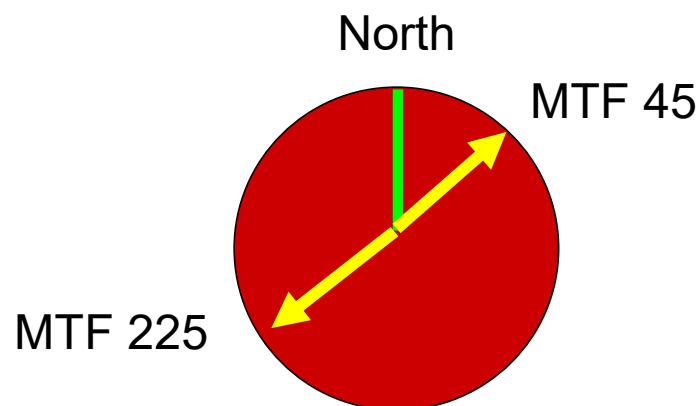


EXECUTION - Steering the Well



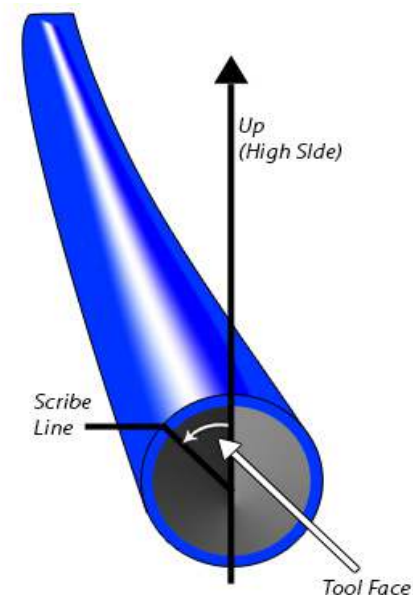
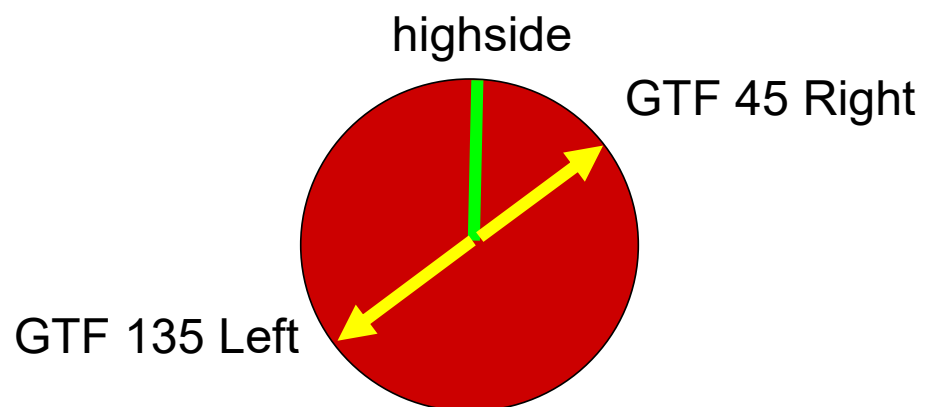
Magnetic Toolface

- Referenced to the north
- When steering in vertical / near-vertical section
- Looking down the borehole, MTF is defined as compass degree



Gravity Toolface

- Referenced to the 'highside' of the hole When steering in inclined section
- Looking down the borehole, GTF is defined as degree Right / Left



EXECUTION - Magnetic or Gravity Toolface

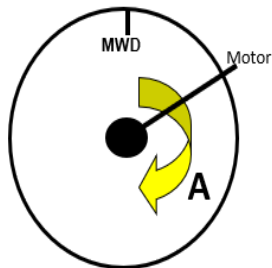
- How do you determine if you are using magnetic or gravity toolface?
 - DDU shows what toolface reference is being used
 - If inclination is $< 5^\circ$ = magnetic, if inclination is $> 5^\circ$ = gravity (according to standard; can be changed)
- Where do special considerations need to be taken when determining the toolface crossover setting?
 - Based on the standard and directional drilling needs
 - If sensor is inside casing when kicking off
 - Depending on well profile
 - In low inclination tangent sections, it may be necessary to deviate from default crossover setting of 5°

Rig Floor Offset

Rig Floor Offset Calculation

RFO Visual Estimate : 60°

Looking Down



RFO Circumference Method

A = (Clockwise looking downhole) MWD scribe to Motor bend

B = Circumference

$$\text{RFO} = (A) \frac{111.00}{675.00} \times 360 = 59^\circ$$

Sperry Drilling

Offset Procedure Checklist

Initial

- ☐ All probe connections are torqued and an end-to-end torque is completed prior to conducting the HSO
- ☐ The scribe line of the hang off collar is aligned with the keyway inside the collar
- ☐ The scribe line is clearly marked and there is only one scribe line
- ☐ The on-tour DD and MWD engineer must both be present when the HSG is determined
- ☐ The on tour DD and MWD engineer independently verify that the BHA is properly torqued
- ☐ The on tour DD and MWD engineer independently measure and verify the RFO
- ☐ Wherever possible, the RFO measurement is performed at the mud motor bend so that both the MWD scribe line and the mud motor reference can be visually verified
- ☐ The on tour DD and MWD engineer are both present when the HSO and RFO are entered into INSITE
- ☐ This sheet and a screen capture of INSITE tool parameters and a screen capture of INSITE location and survey information are sent to the coordinator or ROC for confirmation before drilling commences
- ☐ During the shallow pulse test verify that the correct RFO is displayed as the correction on the session log for the Toolface data list
- ☐ When using a surface readout gyro, the on-tour MWD and DD engineer both ensure that the UBHO mule-shoe key is lined up with the high-side of the motor or whipstock and ensure the mule-shoe sleeve is properly locked in place

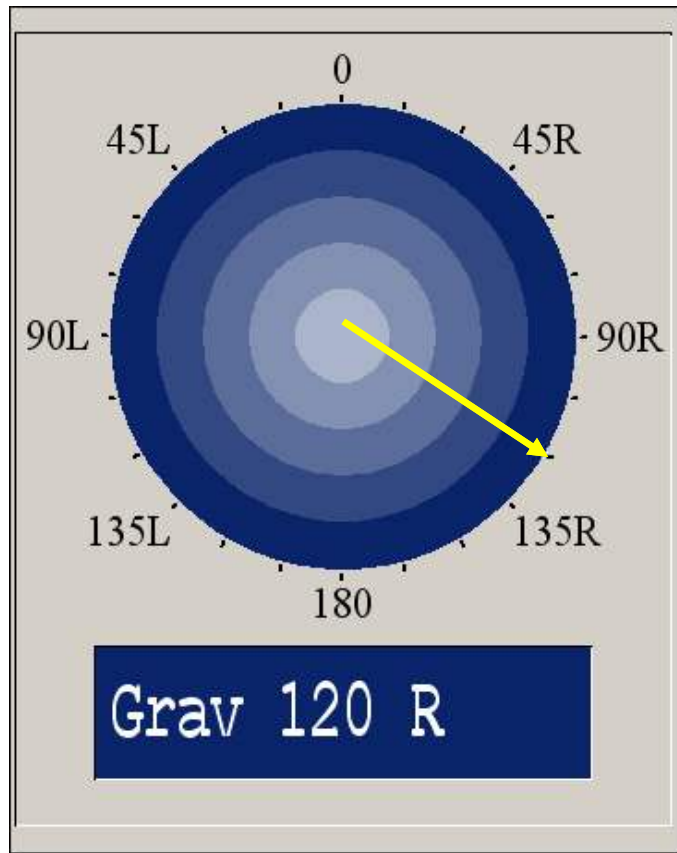
Signed: _____

Orienting Toolface

- Prior to orienting the toolface, record:
 - Pickup weight
 - Slackoff weight
 - Rotating weight off bottom
 - Stand pipe pressure off bottom

- Instruction to give the driller:
 - Work the drill string up and down to release any trapped torque
 - DD to ensure consistent toolface reading after working the drill string
 - Scribe the drill pipe for a visual reference when orienting the toolface
 - DD needs to verify if using magnetic or gravity toolface
 - Turn the drill string left or right to orient the toolface (magnetic toolface or gravity toolface) to desired value taking into consideration the amount of anticipated reactive torque
 - Work the drill string again to work the torque down the string
 - DD to ensure consistent toolface reading again before beginning to drill

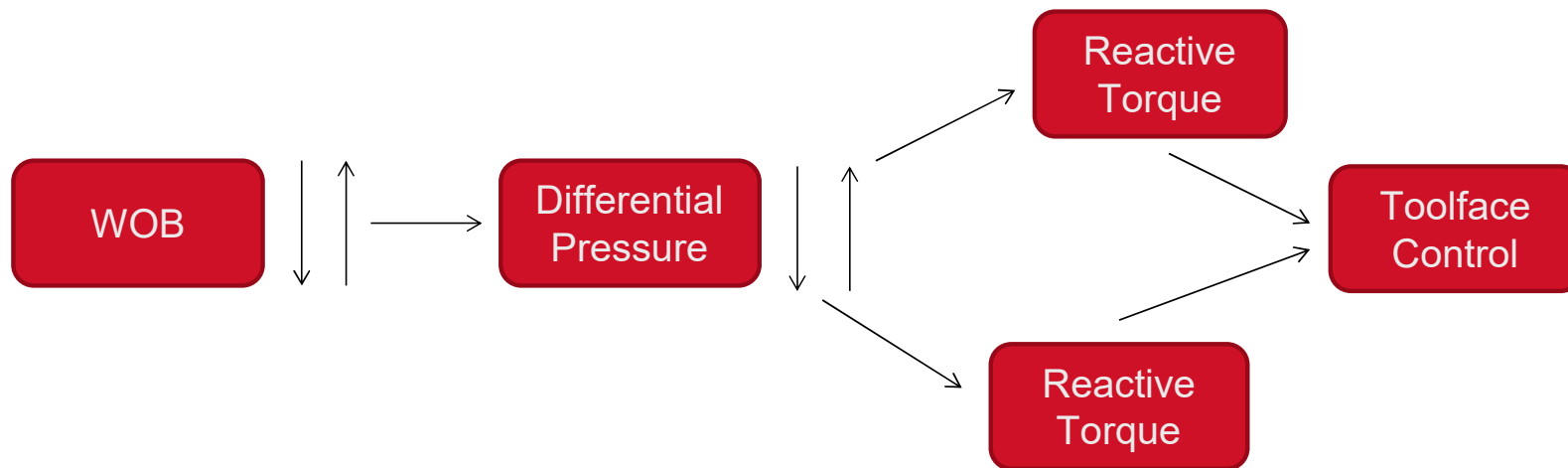
Driller's Display Unit



- Displaying Toolface update from the MWD downhole on the Rig Floor
- Reference for the Directional Driller to orient the drillpipe

EXECUTION - Steering Efficiency

- How does reactive torque affect toolface control?
 - The downhole motor is subject to reactive torque which is the tendency of the drill string to turn in a direction opposite of the bit resulting in torque of the drill string as pressure is applied to the bit



EXECUTION - Sliding

- What is the purpose of sliding?
 - To deviate the well from point A to point B in terms of inclination and azimuth
- What does it mean to slide?
 - Drilling the wellbore with the downhole motor oriented in a specific toolface (magnetic or gravity) with the main purpose of deviating the well
 - The drill string is at 0 rotation (only the bit is rotating)

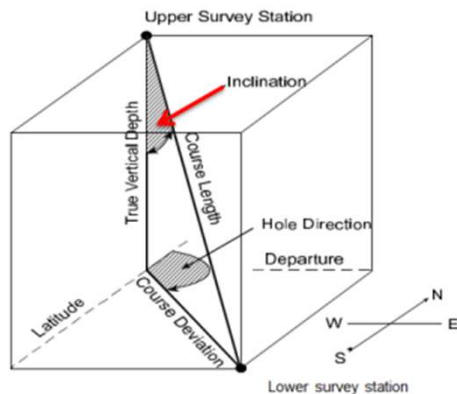
EXECUTION - Reactive Torque

- What factors contribute to the DD's decision making on controlling reactive torque?
 - Understanding the bit profile and design as related to the formation you are drilling through
 - PDC bit, Tri-Cone bit, Natural Diamond bit, etc.
- Other drilling parameters to manage?
 - Flow rate
 - WOB

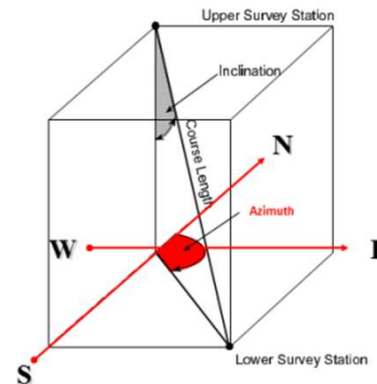
3 Components of Survey

1. Depth
2. Inclination
3. Azimuth

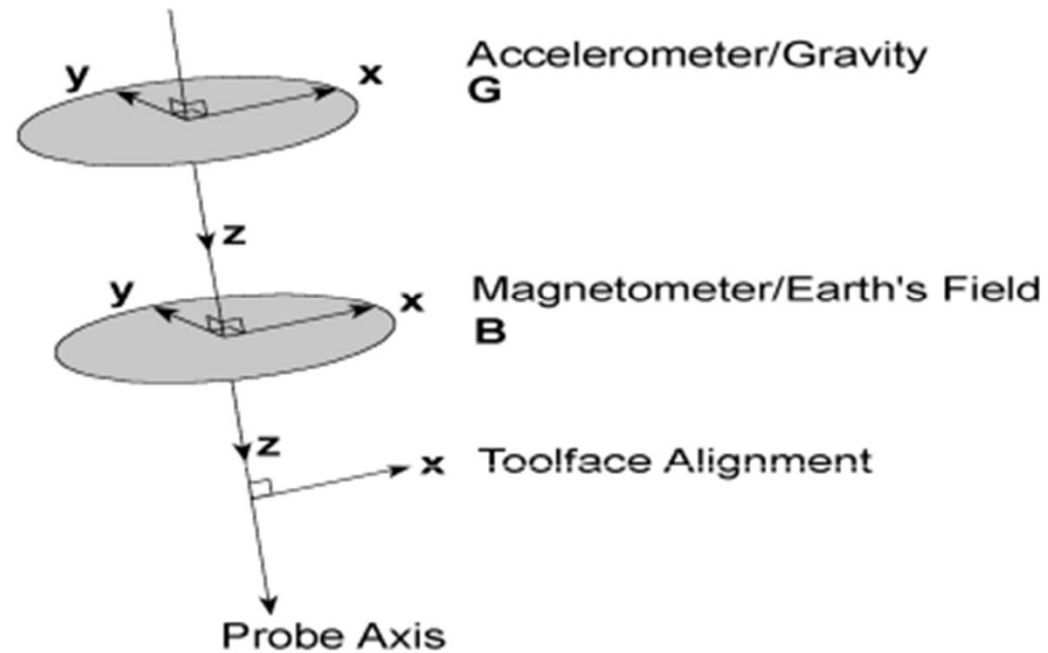
Inclination is the angle, measured in degrees, by which the wellbore or survey instrument axis varies from a true vertical line. An inclination of 0° would be true vertical and an inclination of 90° would be horizontal.



Hole direction (Azimuth) is the angle, measured in degrees, of the horizontal component of the borehole or survey instrument axis from a known north reference. This reference can be: true north, magnetic north, or grid north, and is measured clockwise by convention. Hole direction is measured in degrees and expressed in either azimuth form (0° to 360°) or quadrant form (NE, SE, NW, SW).



Directional Sensor



Inclination - G_x, G_y, G_z

Azimuth - $B_x, B_y, B_z, G_x, G_y, G_z$

Gravity Toolface - G_x, G_y

Magnetic Toolface - $B_x, B_y, B_z, G_x, G_y, G_z$

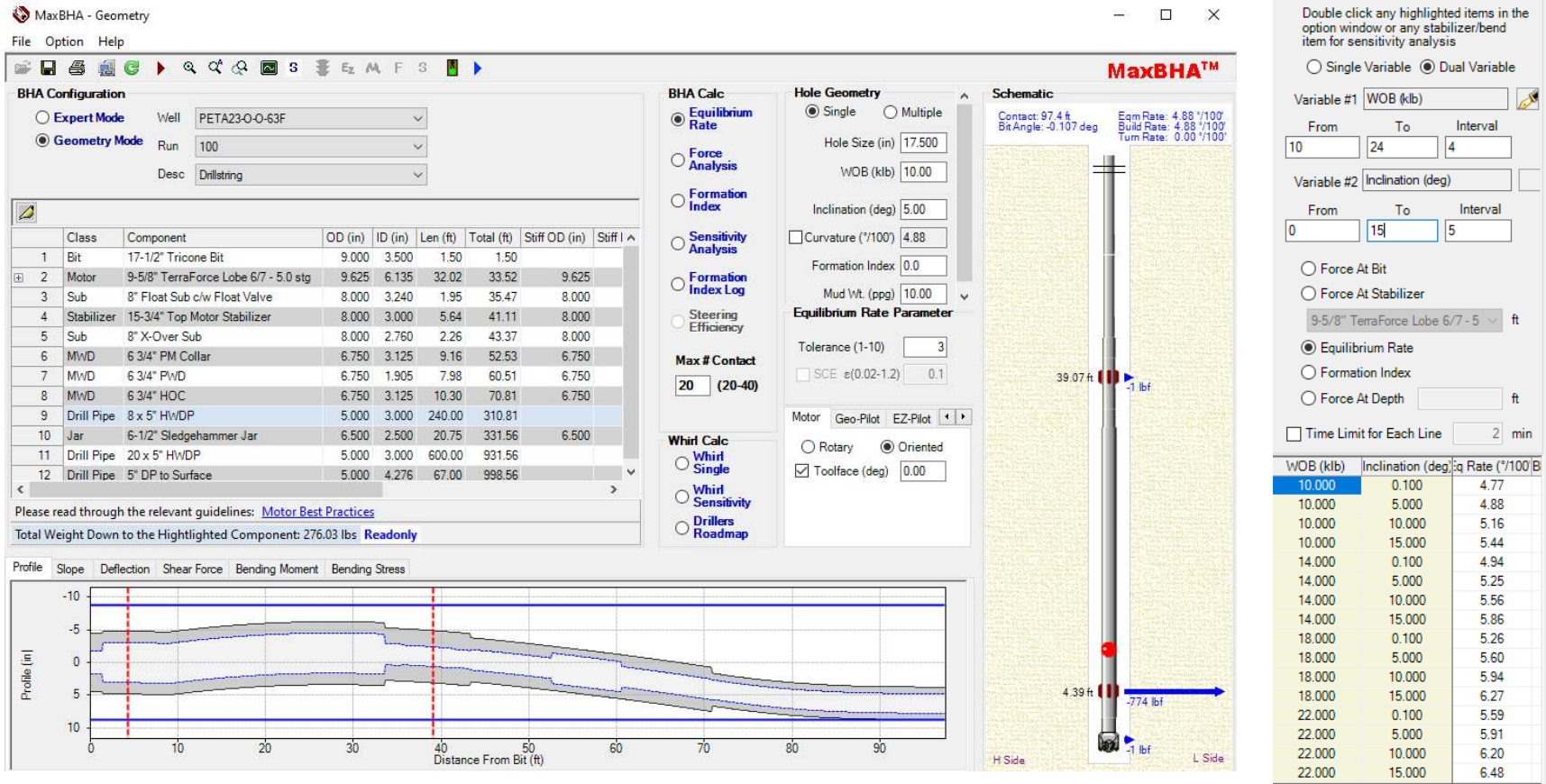
EXECUTION - Survey Frequency and how to verify if a Survey

- Frequency of surveys
 - Based on survey program, A/C requirements and DD's operational needs
- Verify if the survey is successful
 - When G_{total} ($\pm 0.003G$), B_{calc} (± 300 nT), and dip angle (± 0.3 deg) is within acceptable criteria
 - When it falls within the limits calculated by the DD
- DD receives the survey data once the MWD receives and accepts the data in INSITE
- DD must record the survey data into COMPASS and double check the data with MWD

EXECUTION - Achieving Dogleg

- How is the required Dogleg from the well plan achieved?
 - Downhole Motor
 - Slide based on predicted motor yield at required toolface (MaxBHA or Motor Book)
 - Adjust slide distance and/or parameters (Flow, WOB, etc.) as necessary to achieve required dogleg

EXECUTION - Achieving Dogleg



EXECUTION - Achieving Dogleg

MaxBHA™

Sensitivity Analysis

Double click any highlighted items in the option window or any stabilizer/bend item for sensitivity analysis

☐ Single Variable ☒ Dual Variable

Variable #1 WOB (klb)

From 10 To 24 Interval 4

Variable #2 Inclination (deg)

From 0 To 15 Interval 5

☐ Force At Bit
☐ Force At Stabilizer
☒ Equilibrium Rate
☐ Formation Index
☐ Force At Depth

9-5/8" TerraForce Lobe 6/7 - 5 ft

☐ Time Limit for Each Line 2 min

WOB (klb)	Inclination (deg)	q Rate (%/100 B)
10.000	0.100	4.77
10.000	5.000	4.88
10.000	10.000	5.16
10.000	15.000	5.44
14.000	0.100	4.94
14.000	5.000	5.25
14.000	10.000	5.56
14.000	15.000	5.86
18.000	0.100	5.26
18.000	5.000	5.60
18.000	10.000	5.94
18.000	15.000	6.27
22.000	0.100	5.59
22.000	5.000	5.91
22.000	10.000	6.20
22.000	15.000	6.48

Current Inc	5	Deg
Motor Output	4.88	Deg
DLS Required	2.5	Deg
Course Length	100	ft
Slide Required	(DLS required/Motor Output)* Course Length	
	(2.5/4.88)*100	
	51.2	ft

DAILY PROGRESS				Date : 4-Nov-23		Well Name		PETA23-P07 O-G-03F		Depth In (ftMD)		90 ftMD		LAST SURVEY DATA								HALLIBURTON Sperry Drilling Services DRILLING PARAMETERS AND SURVEY SHEET																																							
Dist		%		Time(hrs)		ROP(ft/hr)		Field Name		Petani		Depth Out (ftMD)		998 ftMD		Sey Depth		998.00		Distance to Plan																																									
SLIDE		483		53.2%		2.93 Hrs		164.66		RIG		ADM-88		Drill Hrs		6.23 Hrs		Inclination		16.93 deg						Ct - Ct		9.30		TF		-162																													
ROTATE		425		46.8%		3.30 Hrs		128.79		BHA Type		9-5/8" Mud Motor, ABH 1.15'		Total Slide(ft)		483 (53.2%)		Azimuth		329.18 deg						Vertical		8.80 Ft		Above		Flight																													
Tot. Drilled		908		ROP AVG		145.67 ft/hr		Hole Size		17-1/2in		Total Drilled (ft)		908 ft				TVD		980.30		Horizontal		2.90 Ft		Flight																																			
Date		Time		Depth		Drilling Mode		Toolface		Slide Dist		Rotate Drilled		ROP ft/hr		Drill Time (Hrs)		RPM		VCB (Kbbl)		Flow Rate		Off Btm Press		On Btm Press		PIU Weight		SID Weight		Rox Weight		Drilling Torque On/Off(Kbbl-ft)		DH Temp (deef)		No		Depth		Inc		Asm		TVD		Dls		BUR		Turn		Drct Dist		Remarks					
4-Nov-23		6:25		6:37		90		127		Rotary				37		185.0		0.20		40		2-4		470		450		550		24		24		24		0.5-0.9																									
		7:16		7:18		127		134		Rotary				7		210.0		0.03		40		4-6		470		450		550		24		24																KOP @135 ftMD													
		7:19		7:31		134		179		Oriented		326M		45				225.0		0.20		40		4-6		470		450		550								94		2		140.20		0.28		279.40		140.20		0.20		0.20		0.00		0.30					
		7:32		7:34		179		188		Rotary				9		270.0		0.03		40		4-6		470		500		600		26		26		26		0.8-1.3																									
		8:02		8:11		188		204		Rotary				16		106.7		0.15		40		4-6		470		500		600																																	
		8:12		8:24		204		249		Oriented		326M		45				225.0		0.20		40		4-6		470		500		600		27		27		27		0.85-1.4		94		3		201.00		0.97		313.68		201.00		1.24		1.13		56.38		0.50			
		8:24		8:52		249		272		Rotary				23		49.3		0.47		50		4-6		500		600		700										96.9		4		261.40		1.76		331.67		261.40		1.47		1.31		23.78		0.70					
		8:56		9:06		272		310		Oriented		326M		38				228.0		0.17		40		4-6		500		600		700		30		30		30		0.9-1.45																							
		9:35		9:36		310		315		Rotary				5		300.0		0.02		50		8-10		500		600		700																																	
		9:41		9:54		315		372		Oriented		326M		57				263.1		0.22		80		8-10		500		600		700		33		33		33		1-1.5		99.9		5		343.20		4.74		323.27		343.00		3.88		3.64		-10.27		1.40		Hole, PIU and MU Jar	
		11:03		11:10		372		391		Oriented		HS		19				162.9		0.12		80		8-10		500		600		700																															
		11:49		11:52		391		407		Rotary				16		320.0		0.05		60		8-10		550		800		900										102.8		6		402.70		6.20		321.98		402.30		2.46		2.45		-2.17		1.70					
		11:55		12:11		407		451		Oriented		10R		44				165.0		0.27		80		8-10		550		800		900		36		36		36		1.1-1.6																							
		12:44		12:48		451		460		Rotary				9		135.0		0.07		60		8-10		550		800		900																																	
		12:57		13:18		460		512		Oriented		10R		52				148.6		0.35		80		8-12		550		800		900		38		38		38		1.2-1.7		105.8		7		463.60		8.22		321.23		462.70		3.32		3.32		-1.23		2.10		Switch MP 1to MP 3	
		13:51		13:54		512		522		Rotary				10		200.0		0.05		60		8-10		550		850		950																																	
		13:59		14:24		522		572		Oriented		30R		50				120.0		0.42		80		8-14		550		850		950		41		41		41		1.3-1.8		105.80		8		523.90		9.81		328.80		522.20		3.28		2.64		12.95		2.10		Adjunt RPM TDS	
		15:05		15:10		572		579		Rotary				7		84.0		0.08		60		8-10		550		900		1000																																	
		15:12		15:34		579		633		Oriented		HS		54				147.3		0.37		80		8-14		550		900		1000		43		43		43		1.4-1.85		108.70		9		585.10		11.88		330.37		582.30		3.42		3.38		2.57		1.30			
		16:10		16:12		633		640		Rotary				7		210.0		0.03		60		8-10		550		900		1000																																	
		16:15		16:42		640		693		Oriented		HS		53				117.8		0.45		80		8-14		600		1150		1300		47		45		45		1.5-1.9		108.70		10		645.20		14.32		329.53		640.30		4.07		4.06		-1.40		1.50		Hole	
		17:39		18:00		693		754		Rotary				61		174.3		0.35		60		8-12		600		1150		1300										111.70		11		706.50		14.82		328.34		700.20		0.95		0.82		-1.94		3.00					
		18:39		19:21		754		815		Rotary				61		87.1		0.70		60		8-12		600		1200		1300		51		47		50		1.6-2.0																									
		19:56		20:07		815		841		Oriented		HS		26				141.8		0.18		80		8-12		600		1300		1400																															
		20:08		20:20		841		876		Rotary				35		175.0		0.20		70		10-14		600		1300		1400		54		50		52		1.8-2.1																									
		20:57		21:20		876		937		Rotary				61		159.1		0.38		70		10-14		650		1450		1550										117.60		12		949.70		16.93		329.18		934.10		0.87		0.87		0.35		7.50		Last MWd Survey			
		21:48		22:17		937		998		Rotary				61		126.2		0.48		70		10-14		650		1450		1550		57		53		55		1.9-2.2		117.60		13		998.00		16.93		329.18		990.30		0.00		0.00		0.00		3.30		Projection to CP			

EXECUTION - Achieving Dogleg

MaxBHA™

Sensitivity Analysis

Double click any highlighted items in the option window or any stabilizer/bend item for sensitivity analysis

☐ Single Variable ☒ Dual Variable

Variable #1 WOB (klb)

From 10 To 24 Interval 4

Variable #2 Inclination (deg)

From 0 To 15 Interval 5

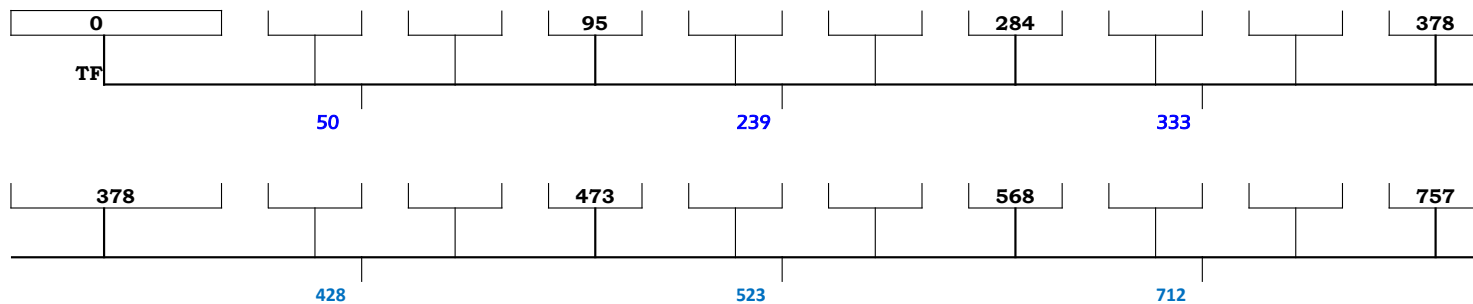
☐ Force At Bit
☐ Force At Stabilizer
 9-5/8" TerraForce Lobe 6/7 - 5 ft
☒ Equilibrium Rate
☐ Formation Index
☐ Force At Depth ft

☐ Time Limit for Each Line 2 min

WOB (klb)	Inclination (deg)	Eq Rate (%/100 ft)
10.000	0.100	4.77
10.000	5.000	4.88
10.000	10.000	5.16
10.000	15.000	5.44
14.000	0.100	4.94
14.000	5.000	5.25
14.000	10.000	5.56
14.000	15.000	5.86
18.000	0.100	5.26
18.000	5.000	5.60
18.000	10.000	5.94
18.000	15.000	6.27
22.000	0.100	5.59
22.000	5.000	5.91
22.000	10.000	6.20
22.000	15.000	6.48

Current Inc	5	Deg
Motor Output	4.88	Deg
DLS Required	2.5	Deg
Course Length	100	ft
Slide Required	(DLS required/Motor Output)* Course Length	
	(2.5/4.88)*100	
	51.2	ft
MWD Offset	45	ft

PIANO SHEET



Questions?

